

# Strategic mistakes in school assignment: Biased beliefs versus optimization errors

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## Abstract

Theory shows that the manipulable Boston mechanism can result in higher expected student welfare than the strategy-proof deferred acceptance (DA) mechanism. Recent empirical studies find, however, the opposite and attribute this to students making strategic mistakes. Such mistakes can be due to biased beliefs about admission probabilities and to optimization errors. We use unique survey data combined with register data from the secondary-school match in Utrecht in 2016 to distinguish these two sources. We find that 18.1% of the students make a strategic mistake, that about two thirds of these mistakes are due to optimization errors instead of biased beliefs, and that optimization errors account for 85% of the associated welfare loss. Informing students about admission probabilities therefore has only a small effect on welfare. Under students' actual choices, mean welfare is lower under Boston than under DA. When all students play optimally, the welfare difference between the two mechanisms disappears but does not reverse. The results favor the strategy-proof DA mechanism. *JEL*-codes: C83; D47; I20.

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# 1 Introduction

In a seminal paper in the school assignment literature, [Abdulkadiroğlu et al. \(2011\)](#) show that the manipulable Boston mechanism may result in higher welfare than the truth-telling deferred acceptance (DA) mechanism because it allows students to express the intensity of their preferences.<sup>1</sup> Subsequent empirical studies by [Agarwal and Somaini \(2018\)](#) and [Calsamiglia et al. \(2020\)](#) corroborate this theoretical prediction in models that assume sophisticated students.

Recent studies by [Kapor et al. \(2020\)](#) and [De Haan et al. \(2023\)](#) find, however, that not all students are sophisticated.<sup>2</sup> Students make strategic mistakes, and in both studies these mistakes reverse the welfare ordering of the mechanisms. [Kapor et al. \(2020\)](#) collect survey data on students' beliefs about admission probabilities and extend the framework of [Agarwal and Somaini \(2018\)](#) with the feature that these beliefs may be biased. They find that high-school applicants in New Haven (Connecticut, US) are too optimistic about admission probabilities in the second round and are therefore not sufficiently strategic. [De Haan et al. \(2023\)](#) use a survey to elicit cardinal preferences for schools of applicants for secondary schools in Amsterdam (the Netherlands). They find that around 8% of the students could have increased their expected utility by making another choice given the choices of the other students. This study cannot distinguish between strategic mistakes due to biased beliefs and strategic mistakes due to optimization errors, whereas the study by Kapor et al. covers only mistakes due to biased beliefs. Disentangling these two sources of mistakes is relevant for school districts that run a manipulable assignment mechanism and want to help participants navigate the mechanism optimally.

In this paper we use unique survey data collected from participants in the secondary-school match in the city of Utrecht in the Netherlands to separate the contributions of biased beliefs and optimization errors as sources of strategic mistakes. In 2016, the school district of Utrecht allowed us to invite participants in the secondary-school match to fill out an online survey concerning school choice. The essential questions in the survey are those that elicit students' beliefs about admission probabilities and their cardinal preferences for schools. To

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<sup>1</sup>[Troyan \(2012\)](#) shows that this welfare advantage of the Boston mechanism extends to settings with priorities when welfare is evaluated ex ante, that is, before students know their own cardinal utilities and priorities. When students do know their own utilities and priorities, neither mechanism needs to dominate the other; see Section 5.3. The Boston mechanism transmits preference intensities indirectly, through equilibrium play. [Eraslan et al. \(2026\)](#) study mechanisms that elicit cardinal utilities directly, and estimate for Seattle that such mechanisms raise mean welfare relative to a strong ordinal mechanism.

<sup>2</sup>[He \(2017\)](#) documents departures from sophistication, finding that parents in Beijing apply too cautiously under the Boston mechanism and rank safe schools too often.

elicit beliefs about admission probabilities we asked students to indicate for each secondary school the probability that they would lose the lottery if applying to it. To elicit cardinal preferences for schools we followed [De Haan et al. \(2023\)](#) and asked students to award points to each school they considered acceptable, relative to their most-preferred school which is normalized at 100 points. The survey thus measures both objects of which [Kapor et al. \(2020\)](#) and [De Haan et al. \(2023\)](#) each observed only one. This allows us to decompose strategic mistakes into the two sources.

For the assignment of students to secondary schools, the school district of Utrecht operates a version of the adaptive Boston mechanism. In the first round students apply to one school. There are a few priority rules and ties are broken by a centralized lottery. Students who remain unassigned in the first round can in the second round only apply to schools that have vacant seats after the first round. We complement the survey data with register data from Utrecht’s secondary-school match. This includes information about students’ actual application choices in Utrecht’s secondary-school match, their priority status, the actual placement and the capacities of schools. With this information we can compute actual admission probabilities.

We merge the data from the school match and the survey at the individual student level. With the merged data, we identify a strategic mistake as an actual school choice that does not maximize the student’s expected utility, given the actual admission probabilities. If the actual choice can be rationalized by the student’s beliefs about admission probabilities, we classify it as a biased-belief mistake. Otherwise the mistake is an optimization error. An optimization error is a residual category: it captures any deviation from expected-utility maximization given the student’s stated preferences and beliefs, ranging from computational errors to coherent motives other than expected-utility maximization. [Section 5.2](#) examines to what extent such non-expected-utility behavior can account for the choices that we classify as optimization errors.

The two categories label the binding source of each mistake, not exclusive types of students. A student who makes an optimization error may have biased beliefs, and vice versa. We find that 18.1% of the students make a strategic mistake. About one third of these mistakes are due to biased beliefs and about two thirds are optimization errors. The mistakes that are due to biased beliefs are mostly made by students who apply to their most-preferred school. The optimization errors fall into two groups. Some students apply to their most-preferred school although a less-preferred school with a higher admission probability would give them higher expected utility. A somewhat larger group applies to a school other than their most-preferred school, but chooses the wrong alternative. Half of all mistake-makers

forgo less than 5 preference points, but the losses in the top decile exceed 28 points. Optimization errors account for 85% of the total welfare loss.

Strategic mistakes are costly and determine the welfare comparison of the two mechanisms. Under students' actual choices, mean welfare is lower under Boston than under DA, where students are assumed to rank truthfully, which under DA is optimal. When all students instead play their expected-utility-maximizing choice, which is the Nash equilibrium of the application game, Boston closes most of this gap but does not overtake DA. In this Nash equilibrium, mean welfare under Boston is not significantly different from mean welfare under DA. The potential advantage of the Boston mechanism ([Abdulkadiroğlu et al., 2011](#)) thus does not materialize in this market, even when all strategic mistakes are eliminated. Correcting only biased beliefs hardly changes mean welfare, because optimization errors account for most of the welfare loss. This conclusion also holds when we allow the two categories to overlap: the part of the optimization-error losses that biased beliefs can explain is small (Section [5.2](#)).

Measuring preferences and beliefs directly also indicates why optimal-choices Boston does not outperform DA in the market that we study. The Boston mechanism will perform better when it allocates the scarce seats to the students who gain most from a seat relative to their fallback after a rejection. In our data a safer application often yields little, because the safer school or a close substitute is still open in the second round, so most students apply to the oversubscribed school. The students who do apply to a safer school are those with a valuable safe alternative, not those with the smallest gain from a seat. Also, a rejected student loses access to schools that fill their seats in the first round, a cost that DA avoids through deferral. Section [5.3](#) elaborates on this.

Our results are related to a growing literature on application mistakes and information policies, reviewed by [Rees-Jones and Shorrer \(2023\)](#). Students submit overly risky lists out of overconfidence about their admission probabilities ([Arteaga et al., 2022](#)), and they make payoff-relevant “obvious” mistakes such as choosing dominated options ([Shorrer and Sóvágó, 2023, 2024](#)). [Larroucau et al. \(2026\)](#) show that a large-scale personalized-information policy substantially reduces such mistakes in Chile's college admissions. Chile assigns students through DA with a constraint on the length of the submitted list. The authors distinguish four sources of mistakes: biased beliefs about admission probabilities, incorrect information about program characteristics, unawareness of available options, and misunderstanding of the mechanism. They also consider non-strategic students, who rank programs by a criterion other than expected utility, such as truthful ranking. This classification maps into our distinction. Their biased-belief mistakes and ours are the same. Their mistakes

from misunderstanding the mechanism and those of non-strategic students are optimization errors in our framework. Mistakes about program characteristics and awareness have no counterpart in our classification, because we take the elicited preference points as the welfare benchmark and the survey covers the full set of schools. Larroucau et al. (2026) find that many students misreport their preferences even when the list-length constraint does not bind, and that part of these misreports is payoff-relevant. Mistakes thus also occur when students have no clear strategic incentive to misreport. Their intervention, which provided personalized information on admission chances and program characteristics, reduced both over-confidence and under-confidence mistakes. Our setting differs in that the mechanism itself is manipulable. Under Boston, most of the welfare loss comes from optimization errors, which information about admission probabilities does not correct.

Our findings are relevant for school districts that operate some version of the Boston mechanism. First, a substantial share of the students make strategic mistakes, which lowers mean welfare below its level under DA. Second, optimization errors account for the larger part of this welfare loss, not biased beliefs. Providing accurate information about admission probabilities will therefore have only a limited impact on welfare. Third, better choices do not restore the case for the Boston mechanism. Even under fully optimal choices its welfare does not exceed that of DA in the market we study.<sup>3</sup>

The paper proceeds as follows. The next section describes the context of secondary-school choice in the city of Utrecht. Section 3 provides more details about the data and presents summary statistics. Section 4 defines strategic mistakes and shows how we decompose them into biased-belief mistakes and optimization errors. Section 5 presents and discusses the results and Section 6 concludes.

## 2 Context

This section briefly describes the structure of secondary education in the Netherlands and secondary school choice in the city of Utrecht.

We study school assignment of students who are finishing primary school and are choosing a secondary school in Utrecht. At this stage, students are 11 or 12 years old. The Netherlands has a tracked secondary-education system, with a four-year vocational track, a five-

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<sup>3</sup>Also, we find no socioeconomic gradient in who makes mistakes. Mistake-makers are no more likely to come from below-median-income households, and the choice of mechanism does not change how students sort across schools by income. The common argument that DA levels the playing field is thus not present in our setting.

year general track and a six-year pre-university track. Track placement is based on students' performance on a nationwide exit test from primary school and the recommendation of primary-school teachers.

Students can freely choose a school that offers the track in which they are placed. Virtually all schools are publicly funded, and there are no tuition fees.<sup>4</sup> Schools are not allowed to select students and can only decline students when the number of applying students exceeds the capacity. In that case the school first applies pre-determined priority rules and then assigns the remaining seats randomly.<sup>5</sup> All schools prepare their students for nationwide exams at the end of secondary education. The Dutch Inspectorate of Education assesses the quality of schools and publishes its findings on the Internet. Schools that receive the lowest quality score for three years in a row lose their public funding and are typically closed.

In this paper we focus on students in the pre-university track in the city of Utrecht. Each year around 1,000 participants in the Utrecht secondary-school match are in the pre-university track, which is about 28% of the cohort. Nine secondary schools in Utrecht offer the pre-university track. Two of these schools exclusively offer this track. We refer to these as single-track schools. The other seven schools are comprehensive schools which also offer the general track and sometimes also the vocational track. All schools give priority to students with an older sibling attending the school and to students with a parent who is employed by the school. The comprehensive schools also give priority to students living in the city of Utrecht. The two single-track schools, the gymnasia, do not give residence priority; they also draw applicants from outside the city. Some students living in Utrecht attend a secondary school in one of the towns surrounding Utrecht (Bilthoven, Breukelen, Houten, Maarssen, Nieuwegein, Woerden, Zeist). In total, 13 schools in these surrounding towns are attended by students living in Utrecht. Two of these schools, Montessori Lyceum Herman Jordan in Zeist and De Werkplaats Kindergemeenschap in Bilthoven, were oversubscribed in 2016 and admitted students by lottery. Applying to them therefore carries admission risk. The school-level tables below report these two schools separately and pool the other schools in the surrounding towns into a single "outside Utrecht" category.

The school district uses a variant of the adaptive Boston (Immediate Acceptance) mechanism for school assignment ([Abdulkadiroğlu et al., 2005](#)). The mechanism consists of two rounds. In the first round, each student applies to one school. Schools rank applicants

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<sup>4</sup>Less than 0.5 percent of Dutch secondary-school students attend a privately-funded school. Publicly-funded schools can request a voluntary contribution from parents. The suggested amounts are so small that financial motives will not influence school choice.

<sup>5</sup>Priority rules are not allowed to involve academic achievement.

in order of their priority and break ties by lottery numbers. Oversubscribed schools reject the lowest-ranked students. A school can also, in consultation with the school district, marginally extend its capacity when applications exceed the planned number of seats. In 2016 two schools (St. Bonifatius College and Utrechts Stedelijk Gymnasium) extended their capacity and thereby avoided a lottery. In the second round, the rejected students submit a list of four schools. Officials assign rejected students to schools with empty seats using these lists. Importantly, students assigned in the first round cannot lose their seat to second-round applicants. The schools in the surrounding towns apply decentralized school assignment; each oversubscribed school uses a lottery. Students learn their track in January when they are still in the final year of primary school. The first application round takes place in March and the second in April. The next school year starts in August.

### 3 Data and summary statistics

This section outlines the data that we use in this study. Section 3.1 describes the data sources and explains how our survey elicits students' subjective admission probabilities and their cardinal preferences for schools. Section 3.2 presents summary statistics.

#### 3.1 Data sources

Our data combine register data from the secondary-school match in Utrecht in 2016 and data collected through our online survey. These data are merged at the student level.

**Register data.** The register data contain information from all students who participated in the first round of the secondary-school match in the school district of Utrecht in 2016. The data include the school to which each student applied in the first round, whether the student has priority at that school, the capacities (per track) of each school and the student placement in the first round. The data also contain student characteristics, such as sex, home address, and the score on the nationwide exit test from primary school.

We use the data on applications, priorities and capacities to compute for each school the actual admission probability. Students with priority are ranked above students without priority. In 2016 the number of priority applicants did not exceed the capacity of any school, so all applicants with priority were admitted. Certainty of admission for priority students is thus an outcome of the data. We therefore compute admission probabilities for students

without priority. These probabilities are the means from resampling the full population of applicants in the pre-university track 10,000 times.<sup>6</sup>

We combine information of the home address and the locations of secondary schools to construct distances between each student’s home address and the schools. As a measure of a student’s social background, we use average household income of the neighborhood where the student lives. We look at the level of six-digit postal codes which corresponds to a street or a part thereof. This information is provided by Statistics Netherlands. The score on the nationwide exit test from primary school is normalized to mean zero and standard deviation one within the full sample.

**Survey data.** In March 2016, we conducted an online survey among the students who were in the final year of primary school in Utrecht. An invitation to fill out the survey was sent immediately after the application deadline of the first round, and the survey had to be completed before students received the result of the first round. Beliefs and preferences are thus elicited after the round-1 choice was submitted but before its outcome was known.<sup>7</sup> The invitation was sent to the e-mail addresses of the parents, but we refer to respondents as students. The goal of the survey was to elicit students’ cardinal preferences for secondary schools and their subjective admission probabilities for each school.

To elicit students’ cardinal preferences for schools we developed an online version of the survey questions of [De Haan et al. \(2023\)](#). The survey first asked students which school they prefer most. It explained that this does not need to be the school the student applied to in the first round. Students were also asked to indicate which other schools they consider acceptable. Schools they considered unacceptable could be dragged to a separate block.

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<sup>6</sup>In each resample we draw, with replacement, a population of first-round applicants of the same size as the observed one. Within a school, applicants with sibling priority are seated first and the remaining seats are allocated by lottery among the applicants without priority. This implies an admission probability of  $\min\{1, (c_s - p_s)/n_s\}$  at school  $s$  in a given resample, where  $c_s$  is the school’s capacity,  $p_s$  the number of resampled applicants with priority, and  $n_s$  the number of resampled applicants without priority. The reported admission probability is the average of these rates across the resamples. It is an *ex ante* probability because it integrates over the random composition of the applicant pool. It is the admission probability an applicant faces before the set of competing applicants and the lottery outcome are realized. The resampling uses all applicants, not only the survey respondents, because admission probabilities reflect competition from the full applicant pool. For the two gymnasia, Christelijk Gymnasium Utrecht and Utrechts Stedelijk Gymnasium, the applicant pools additionally include the out-of-district applicants who are not in our register data.

<sup>7</sup>Eliciting beliefs after the choice raises the concern that respondents rationalize the choice they just made, for instance by overstating the admission probability at the school they chose. The design and the data limit this concern. Beliefs are elicited for every school on the student’s list rather than only the chosen one, no admission outcomes were known yet, and the dominant pattern in the responses is pessimism about admission probabilities across the board (Section 3.2), which is the opposite of what self-serving rationalization would produce.

Next, the survey asked students to assign preference points to each school. The students' most-preferred school automatically received 100 points. Specifically, the survey posed the following question (translated from Dutch) to elicit preference points on a scale from 0 to 100:

*"For each school we ask you to award points. Your most-preferred school receives 100 points. The points you give to another school, can be seen as a percentage of your most-preferred school. The lower a school is on your list, the fewer points you award. If a school on the list is very close to the previous one, the difference in points is small. For clarity we give some examples."*

Two numerical examples illustrate the concept of preference points. One example describes a student who values a school on her list at 60% of her top-ranked school and therefore gives this school 60 points. The other example mentions: "Student 1 likes school A just as much as school B and likes school C a bit, but does not want to go to any of the other schools. That is why student 1 gives schools A and B both 100 points and school C receives 5 points. The other schools receive 0 points and therefore do not need to be filled in." This implies that students are considered to be indifferent between all unranked schools and that the value of an offer from such a school is normalized to zero. [De Haan et al. \(2023\)](#) show that this measure implies that a student who assigns  $x$  points to a school is indifferent between an offer from that school and a lottery where the student receives an offer from her top-ranked school with probability  $x/100$  and an offer from an unranked school with probability  $(1 - x/100)$ . The preference points thus place each student's schools on a common 0–100 scale. What this scale implies for comparing welfare across students is discussed where we use it, in Section 5.3.

To elicit students' subjective admission probabilities the survey includes the following question:

*"Can you, for each secondary school, indicate what you think the probability is that your child loses the admission lottery if your child would have applied to that school?"*

The survey provided a list of schools offering the student's track and explained that the requested probability equals 0 if the respondent expects no lottery.

### 3.2 Summary statistics

**Students.** Table 1 shows the sample means and standard deviations of student characteristics separately for respondents and non-respondents to the survey. Out of the 950 stu-

dents in the pre-university track who participated in Utrecht’s secondary-school match in 2016, 431 (45.4%) completed the survey. The  $p$ -values reveal no significant differences in observed characteristics between respondents and non-respondents. The share of female students among respondents and non-respondents is 48% and 49%, respectively. The average standardized test score of the non-respondents exceeds the respondents’ test score by 2% of a standard deviation. The share of students with below median household income is 15% among non-respondents, which is 3 percentage points higher than the share among respondents. A joint test indicates that student characteristics do not predict survey-response ( $p$ -value = 0.14).

Table 1: Summary statistics: student characteristics

|                         | Non-respondents | Respondents     | $p$ -value |
|-------------------------|-----------------|-----------------|------------|
|                         | (1)             | (2)             | (3)        |
| Female                  | 0.49<br>(0.50)  | 0.48<br>(0.50)  | 0.98       |
| Standardized test-score | 0.01<br>(1.00)  | −0.01<br>(1.00) | 0.76       |
| Income below median     | 0.15<br>(0.36)  | 0.12<br>(0.32)  | 0.12       |
| # Students              | 519             | 431             |            |

*Notes:* The sample is all 950 participants in the pre-university track of the 2016 match, split into survey respondents and non-respondents. Standard deviations in parentheses. Each  $p$ -value in column (3) comes from a separate regression of the characteristic on a survey-response indicator. The standardized test-score is missing for 4% (3%) of the respondents (non-respondents). Income is missing for 8% (8%) of the respondents (non-respondents).

Table 2 shows how many students in the pre-university track apply to each school, both among all 950 applicants (column 1) and among the 431 survey respondents (column 2). The school with the most applicants, in both samples, is Christelijk Gymnasium Utrecht, which is one of the two single-track schools. The other single-track school is Utrechts Stedelijk Gymnasium. The remaining columns summarize the survey respondents who applied to each school. Column (3) shows that, on average, the students applying to Amadeus Lyceum live about 2 kilometers from that school. Traveling distances to schools outside Utrecht are often larger. The share of applicants with an older sibling in the school varies across schools. Ignoring schools with fewer than five responding applicants, it lies between 5% and 37%.<sup>8</sup>

<sup>8</sup>The sibling indicator in this column comes from the survey and equals one if the student reports having an older sibling enrolled at the school. We use this survey measure only descriptively, here and in Table 3, because it is available for every school a student ranks. All analyses use the sibling priority variable from the administrative records, which is observed at the school of application.

Column (5) reports per school the share of respondents for whom the school where they applied was their most-preferred school. Averaging this column over the respondents shows that about one in nine students (11%) applied to a school other than their most-preferred one. Most students thus apply to their most-preferred school, and strategic behavior is concentrated at a few schools. For example, only 69% of the 42 students applying to Leidsche Rijn College report that this is their most-preferred school. The final column reports each school's ex ante admission probability, computed as described in Section 3.1. The two schools in Utrecht that were oversubscribed and had a lottery were Christelijk Gymnasium Utrecht and Gerrit Rietveld College. In both schools about 20% of the applicants were admitted with priority, because they had an older sibling in the school. The admission probabilities for the students without priority were 0.69 and 0.81, respectively. As expected, the schools with the lowest admission probabilities have the highest shares of applicants for whom the school is the most-preferred.

Table 2: School of application

| Schools                                         | Applicants<br>(all)<br>(1) | Applicants<br>(survey)<br>(2) | Distance<br>(km)<br>(3) | Sibling<br>(4) | Most-<br>preferred<br>(5) | Admission<br>probability<br>(6) |
|-------------------------------------------------|----------------------------|-------------------------------|-------------------------|----------------|---------------------------|---------------------------------|
| Amadeus Lyceum                                  | 69                         | 28                            | 1.99                    | 0.18           | 0.89                      | 1.00                            |
| Christelijk Gymnasium Utrecht <sup>*,+,**</sup> | 190                        | 95                            | 3.17                    | 0.21           | 0.97                      | 0.69                            |
| De Passie                                       | 10                         | 3                             | 4.65                    | 0.67           | 0.67                      | 1.00                            |
| Gerrit Rietveld College <sup>*,+</sup>          | 110                        | 63                            | 2.87                    | 0.16           | 0.95                      | 0.81                            |
| Leidsche Rijn College                           | 108                        | 42                            | 2.33                    | 0.17           | 0.69                      | 1.00                            |
| St. Gregorius College                           | 6                          | 2                             | 6.29                    | 0.00           | 1.00                      | 1.00                            |
| St. Bonifatius College <sup>†</sup>             | 155                        | 63                            | 2.82                    | 0.19           | 0.83                      | 0.97                            |
| UniC                                            | 42                         | 21                            | 3.05                    | 0.05           | 0.71                      | 1.00                            |
| Utrechts Stedelijk Gymnasium <sup>†,**,*</sup>  | 77                         | 38                            | 3.30                    | 0.37           | 0.87                      | 0.97                            |
| Montessori LHJ, Zeist <sup>*</sup>              | 56                         | 19                            | 6.09                    | 0.05           | 0.95                      | 0.85                            |
| De Werkplaats, Bilthoven <sup>*</sup>           | 25                         | 13                            | 6.12                    | 0.08           | 1.00                      | 0.74                            |
| School outside Utrecht                          | 102                        | 44                            | 5.88                    | 0.14           | 0.89                      | 1.00                            |
| Total                                           | 950                        | 431                           |                         |                |                           |                                 |

Notes: Column (1) counts all first-round applicants in the pre-university track; column (2) counts the survey respondents among them. Columns (3)–(5) average over the survey respondents who applied to the school. Column (6) is the ex ante admission probability of an applicant without priority, computed from all applicants (Subsection 3.1). <sup>+</sup>: school with lottery in 2015. <sup>\*</sup>: school with a lottery in 2016. <sup>†</sup>: school that extended its capacity in 2016 to accommodate excess demand (Section 2). <sup>\*\*</sup>: single-track school.

**Non-respondents' application behavior.** Table 2 reports, by school, the number of applicants among all 950 students and among the 431 survey respondents. Comparing the implied shares, non-respondents applied somewhat less often to an oversubscribed school

(62% against 68%,  $p = 0.08$ ) and to the two gymnasia (26% against 31%,  $p = 0.09$ ); the only school-level difference with  $p < 0.05$  is at Gerrit Rietveld College (9% against 15%,  $p = 0.01$ ). If applying to an oversubscribed school makes a strategic mistake more likely, this implies that the share of students making a strategic mistake among respondents may overstate its share in the full population somewhat.

**Stated school rankings.** Ordering the schools by the preference points a student awards them in the survey gives that student’s stated ranking of schools. We interpret this ranking as the rank-ordered list (ROL) that a student would submit to a strategy-proof mechanism, and we use it as such in the counterfactual of Section 5.3, in which students are assigned through deferred acceptance with single tie-breaking (DA-STB). In this counterfactual, each student submits her full stated ranking and all schools break ties by the same single lottery draw per student. Table 3 summarizes the stated rankings, focusing on the five highest-ranked schools. The number of observations declines with the rank because some respondents list fewer than five schools.

Table 3: Summary statistics: stated school rankings

|                           | Rank on ROL      |                  |                  |                  |                  |
|---------------------------|------------------|------------------|------------------|------------------|------------------|
|                           | 1                | 2                | 3                | 4                | 5                |
| Home-school distance (km) | 3.63<br>(2.28)   | 3.74<br>(2.36)   | 4.16<br>(2.39)   | 4.37<br>(2.37)   | 4.66<br>(2.37)   |
| Sibling                   | 0.16<br>(0.36)   | 0.10<br>(0.29)   | 0.04<br>(0.18)   | 0.03<br>(0.15)   | 0.03<br>(0.15)   |
| Single-track school       | 0.33<br>(0.47)   | 0.24<br>(0.42)   | 0.23<br>(0.39)   | 0.26<br>(0.40)   | 0.18<br>(0.34)   |
| Preference points         | 100.00<br>(0.00) | 82.25<br>(16.71) | 65.18<br>(22.19) | 51.90<br>(23.72) | 43.48<br>(23.90) |
| # observations            | 431              | 409              | 387              | 321              | 231              |

*Notes:* The table presents the means of student-school characteristics of the top-5 ranked schools on the stated school rankings of the 431 survey respondents. Standard deviations are in parentheses. The columns indicate the rank on the stated ranking. The sibling indicator is survey-based.

The first row in the table shows that the home-school distance increases in schools’ rank on the ROLs. The mean differences in distance are small. On average, students have several schools to choose from within a 5-kilometer radius of their home. Sixteen percent of the students have a sibling enrolled in their most-preferred school. This declines to 10% for the second-ranked school and to less than 5% for lower-ranked schools. One in three students ranks a single-track school first and one in four ranks such a school second.

The final row of the table shows that students value their second-ranked school on average almost 18% less than their most-preferred school. This gap rises to almost 35% for the third-ranked school and to 48% and 56% for the fourth- and fifth-ranked schools (if ranked). These large differences indicate that it matters on average for students whether they are assigned to their first-ranked school or not.

**Actual and subjective admission probabilities.** Table 4 presents summary statistics on students' subjective beliefs about schools' admission probabilities, next to the actual (rational-expectations) admission probabilities computed from the register data (Section 3.1). Subjective admission probabilities indicate how likely students think it is that they will be admitted to a school if they apply to it.

We treat item non-response on the question eliciting subjective admission beliefs as a belief of certain admission. A school for which a respondent leaves the question blank is assigned a perceived rejection probability of zero, that is, a subjective admission probability of one. This is consistent with the pattern of non-response. Within students' choice sets, the schools left blank are mostly schools that were not oversubscribed. Of the blank school-beliefs, 72% are at schools with an actual admission probability of one, against 35% of the schools for which a belief is given (mean admission probability 0.97 versus 0.90). A separate group of 49 respondents (11%) leave the belief question blank for every school. These respondents apply to oversubscribed schools at the same rate as other respondents (mean admission probability of the applied-to school 0.89 in both groups), so for them the certain-admission rule is a neutral default. We keep these respondents rather than dropping them, so that the subjective-belief benchmark is defined for the entire survey sample. Section 5.4 assesses the robustness of this approach. It re-imputes every missing belief as zero, as certain admission, and as the school's actual admission probability. It also drops the 49 non-reporters altogether. The classification of strategic mistakes and the welfare results are insensitive to the imputation rule.

The comparison of subjective and actual admission probabilities shows two things. Across schools, subjective beliefs are ordered in line with actual admission probabilities: beliefs average above 80% at the schools that were not oversubscribed in 2016 (Amadeus Lyceum, De Passie, Leidsche Rijn College, and UniC), and are lower, between 59% and 76%, at the schools that ran lotteries (Christelijk Gymnasium Utrecht, Gerrit Rietveld College, Montessori Lyceum Herman Jordan, and De Werkplaats Kindergemeenschap) or marginally extended their capacities to accommodate excess demand (St. Bonifatius College and Utrechts Stedelijk Gymnasium). In levels, however, students are systematically too pessimistic: the

mean subjective belief is below the actual admission probability at every school, with the largest gaps at schools that were in fact safe or nearly safe, such as Utrechts Stedelijk Gymnasium (subjective mean 63.5% against an actual probability of 97.4%) and St. Bonifatius College (69.6% against 96.6%).<sup>9</sup>

Table 4: Subjective and actual admission probabilities

| Schools                                          | Subjective beliefs |            | Actual probability |     |
|--------------------------------------------------|--------------------|------------|--------------------|-----|
|                                                  | Mean (%)           | Median (%) | (%)                | N   |
|                                                  | (1)                | (2)        | (3)                | (4) |
| <i>A. Schools in Utrecht</i>                     |                    |            |                    |     |
| Amadeus Lyceum                                   | 84.0               | 93.0       | 100.0              | 141 |
| Christelijk Gymnasium Utrecht*                   | 59.8               | 69.0       | 69.3               | 326 |
| De Passie                                        | 84.1               | 100.0      | 100.0              | 95  |
| Gerrit Rietveld College*                         | 64.2               | 70.0       | 80.8               | 278 |
| Leidsche Rijn College                            | 83.9               | 94.0       | 100.0              | 141 |
| St. Gregorius College                            | 81.1               | 95.0       | 100.0              | 104 |
| St. Bonifatius College <sup>†</sup>              | 69.6               | 80.0       | 96.6               | 241 |
| UniC                                             | 81.2               | 93.5       | 100.0              | 130 |
| Utrechts Stedelijk Gymnasium <sup>†</sup>        | 63.5               | 70.0       | 97.4               | 278 |
| <i>B. Oversubscribed schools outside Utrecht</i> |                    |            |                    |     |
| Montessori Lyceum Herman Jordan, Zeist*          | 76.1               | 80.0       | 84.8               | 18  |
| De Werkplaats Kindergemeenschap, Bilthoven*      | 58.6               | 60.0       | 73.9               | 29  |

Notes: Columns (1) and (2) are computed over the survey respondents who report a belief for the school; column (4) gives their number. Column (3) is the actual admission probability of an applicant without sibling priority, computed from all applicants (as in Table 2). \*: school with a lottery in 2016. <sup>†</sup>: school that extended its capacity in 2016 (Section 2).

## 4 Characterization of strategic mistakes

This section first defines a strategic mistake and then shows how we classify each mistake as due to biased beliefs or to an optimization error. In words, a student makes a strategic mistake when her first-round application does not maximize her expected utility, given her preference points and the actual admission probabilities. A mistake is due to biased beliefs when the application does maximize expected utility under the student's own subjective admission probabilities. Such a student played optimally given her beliefs, and the mistake stems from the beliefs being wrong. A mistake that is suboptimal even under the student's own beliefs is an optimization error.

<sup>9</sup>Figure B1 in the appendix displays the distribution of students' subjective admission probabilities for each school.

We define a strategic mistake from an ex ante perspective: student  $i$ 's first-round application is a mistake when, at the time of application, another school on the student's preference list offered higher expected utility given the actual admission probabilities. Let  $v_{ik}$  denote the preference points that student  $i$  assigns to school  $k$ , and let  $\alpha_{ik}$  denote the actual first-round admission probability of student  $i$  at school  $k$ . These probabilities are equal to one if student  $i$  has priority at school  $k$  and are otherwise equal to the ex ante admission probabilities reported in Table 2. The expected utility of applying to school  $q$  is

$$\text{EU}_i(q) = \alpha_{iq} \cdot v_{iq} + (1 - \alpha_{iq}) \cdot v_{i,\text{out}}, \quad (1)$$

where the outside option  $v_{i,\text{out}}$  is the utility student  $i$  secures in the second round if she is rejected (defined below). The school the student applied to,  $s_i^*$ , is not a strategic mistake if and only if

$$\text{EU}_i(s_i^*) \geq \max_q \text{EU}_i(q), \quad (2)$$

where  $q$  runs over the schools on student  $i$ 's preference list. Restricting  $q$  to the preference list is without loss of generality, because an unranked school has a value of zero, so applying to it is weakly dominated by applying to the most-preferred school. Dividing both sides of (2) by  $\alpha_{is_i^*}$ , the admission probability at the chosen school, and rearranging gives the equivalent threshold condition

$$v_{is_i^*} \geq \max_q \left( \frac{\alpha_{iq}}{\alpha_{is_i^*}} \cdot v_{iq} + \left( 1 - \frac{\alpha_{iq}}{\alpha_{is_i^*}} \right) \cdot \underbrace{v_{i,\text{out}}}_{\text{Outside option}} \right),$$

which we use in the graphical representation of the classification (Section 5.1 and Appendix C).<sup>10</sup>

The outside option  $v_{i,\text{out}}$  is the student's utility if she is not admitted to the school she applies to and goes to the second round. We model the second round explicitly: a student who is rejected re-applies in a simple one-shot Boston round. Appendix A.2 defines the expected utility  $\text{EU}(s)$  of applying to school  $s$  in this round (equation (A1)), using the second-round admission probabilities of Table A1. This expected utility includes what happens after a second rejection: the student is then placed in the best safe school they ranked. The value of the second round is the expected utility of the optimal second-round application,

<sup>10</sup>When  $\alpha_{iq} > \alpha_{is_i^*}$ , the weight on the outside option in the threshold condition is negative. This is innocuous, because the condition is an algebraic rearrangement of the expected-utility comparison in (2), and the classification evaluates (2) directly.

$v_i^{r2} = \max_s EU(s)$ , where  $s$  runs over the ranked schools with vacant seats in the second round. We set  $v_{i,out} = v_i^{r2}$ . Students who ranked no school with seats in the second round obtain  $v_{i,out} = 0$ .

Two features of this construction deserve comment. First, schools outside Utrecht are part of the analysis: the survey elicited preference points for schools in the surrounding towns, which we pool into a composite “outside Utrecht” alternative. This alternative is available in the second round without rationing, so a student who ranks a school outside Utrecht never falls below its value. Appendix E assesses the robustness of the classification to this assumption: it reruns the classification with the outside-Utrecht alternative removed from the second round and with its value cut in half. Second,  $v_{i,out}$  does not depend on the school the student applies to in the first round. The seats that remain for the second round are determined by the first-round applications of the full applicant pool, and no single student’s application measurably changes this residual supply. The welfare simulations of Section 5.3 do re-run the second-round match in every draw, so there the residual seats do respond to the counterfactual first-round choices.

To classify each strategic mistake, we substitute the student’s subjective admission probabilities into condition (2).<sup>11</sup> If the condition holds under the subjective probabilities, we classify the mistake as biased beliefs; otherwise as an optimization error. This procedure is conservative. A student whose choice happens to be consistent with their stated beliefs is classified as belief-driven even if they did not actively maximize expected utility, whereas a purely belief-driven mistake always satisfies condition (2) under the student’s own beliefs. The biased-belief count is therefore an upper bound on the mistakes for which beliefs are the sole cause, and the optimization-error count a lower bound. Our finding that optimization errors are the larger source therefore understates rather than overstates their importance.

The term “optimization error” is a residual category. It covers every deviation from risk-neutral expected-utility maximization at the stated preferences and stated beliefs: computational mistakes, misunderstanding of the mechanism, including the belief that one should always report one’s true first choice (Chen and Sönmez, 2006; Pathak and Sönmez, 2008), regret aversion (Loomes and Sugden, 1982), a direct utility value of winning or losing the lottery, probability weighting (Prelec, 1998), and noise in the elicited preference points all end up under this label. Our data do not separate these channels, and not every one of them is a mistake in a normative sense. The classification establishes that the choice is not rational-

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<sup>11</sup>Because the survey elicits admission beliefs for the first round only, this substitution replaces the first-round admission probabilities, while the value of the second round keeps its modeled value on both sides of the comparison.

izable by the stated preferences and beliefs within the expected-utility benchmark. Section 5.2 tests three non-expected-utility models directly.

## 5 Results

This section presents the results. Section 5.1 reports the incidence and sources of strategic mistakes, Section 5.2 who makes them and what they cost, Section 5.3 the welfare comparison of Boston and DA-STB, and Section 5.4 robustness checks.

### 5.1 Incidence and sources of strategic mistakes

Figure 1 plots all students who made a strategic mistake by the school they applied to on the vertical axis, the school they should optimally have applied to on the horizontal axes, and whether their mistake is due to biased beliefs (in the left-hand graph) or to an optimization error (in the right-hand graph).

The graphs show that 78 of the 431 students (18.1%) make a strategic mistake. Of these 78 mistakes, 29 (37%) can be classified as biased beliefs. The other 49 (63%) are due to optimization errors.<sup>12</sup> The largest cluster of mistakes consists of the 20 students who applied to Christelijk Gymnasium Utrecht (CGU) whereas applying to Utrechts Stedelijk Gymnasium (USG) would have maximized their expected utility. For 11 of these 20 students, the mistake is classified as biased admission beliefs. These students underestimated the probability of being admitted to USG. The other mistakes are scattered across different combinations of actual and optimal choices.

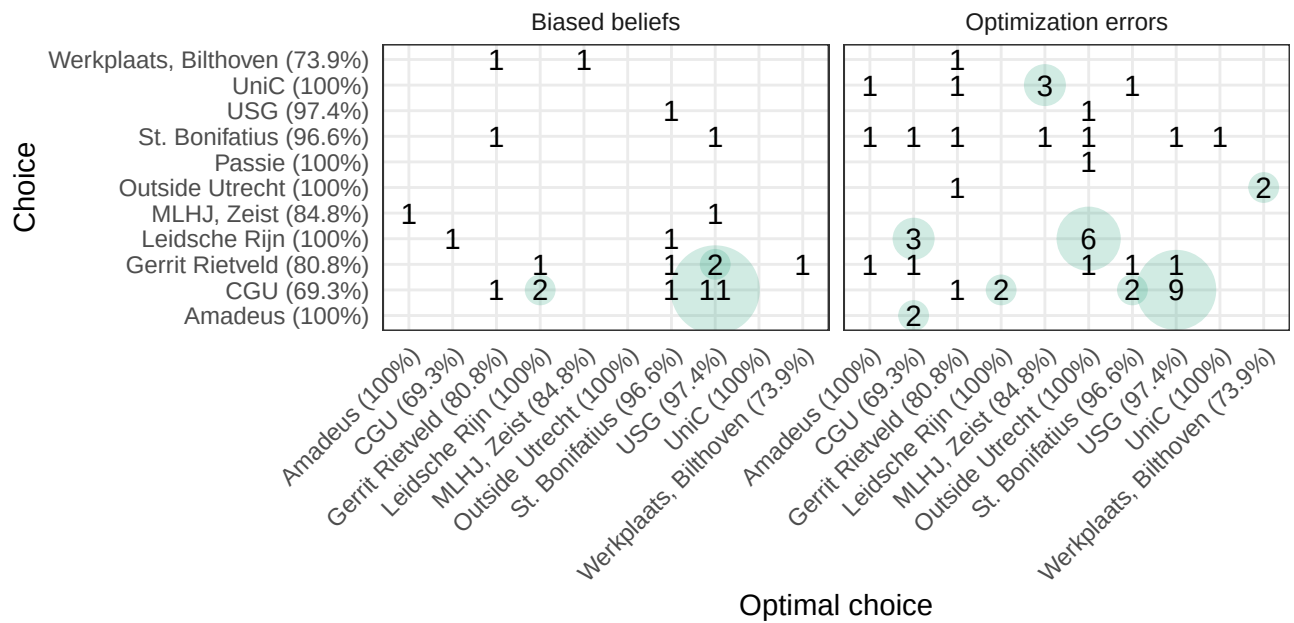
Figure 2 plots the actual choices of the students who made a strategic mistake against the school they prefer most. The left-hand graph shows that of the 29 students who make a mistake due to biased beliefs, 23 applied to the school they prefer most. These students are sincere (not revealed strategic). The right-hand graph shows that of the 49 students who make an optimization error, 16 are sincere and the other 33 made a strategic choice.<sup>13</sup>

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<sup>12</sup>The reverse pattern also occurs. For 24 students (5.6%), the actual choice maximizes expected utility under the actual admission probabilities but not under their own beliefs. We do not count these choices as mistakes. Twenty-three of these students apply to their most-preferred school, and 23 underestimate their admission probability at the school they apply to. Their median subjective admission probability is 49%, against a median actual admission probability of 83%. Seven of them have sibling priority and are certain to be admitted. Under their own beliefs, a safer school would have given them a higher expected utility, by a median of 6.4 preference points. For these students, biased beliefs and a failure to optimize offset each other.

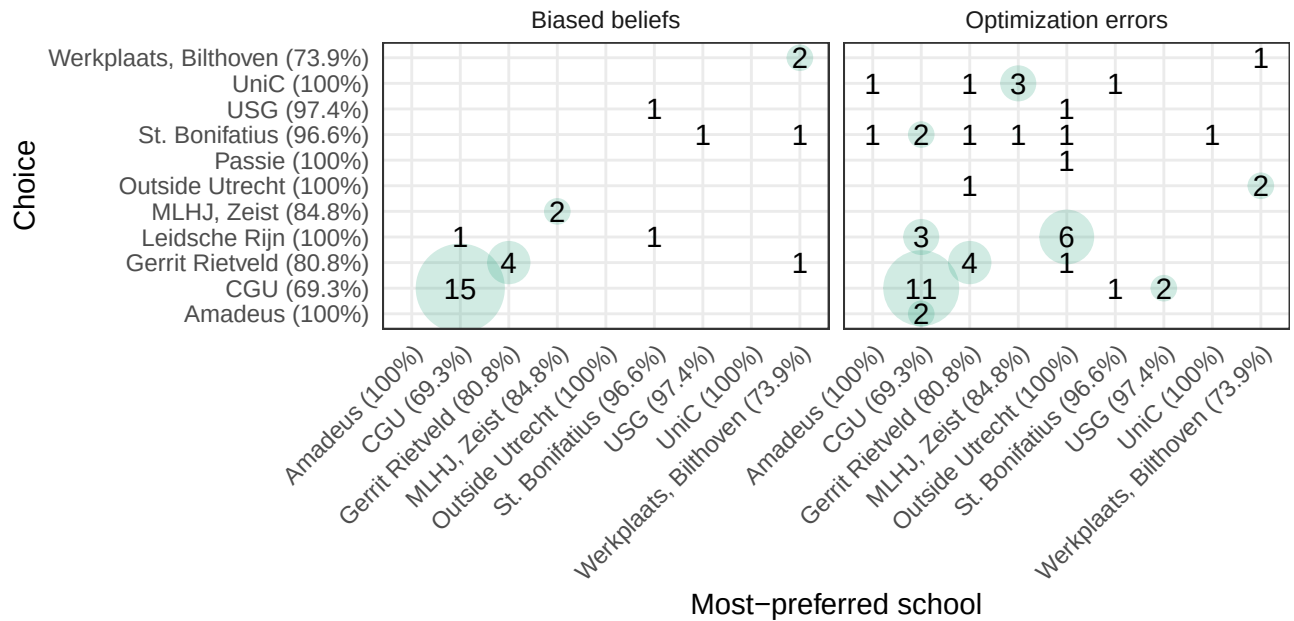
<sup>13</sup>The results in Figures 1 and 2 are based on ex ante admission probabilities, which average over resampled applicant pools. Interim admission probabilities instead condition on the realized applicant pool (see Section

Figure 1: Strategic mistakes: choice vs. optimal choice



Notes: The graphs plot all students who made a strategic mistake. On the vertical axis are the schools they actually chose and on the horizontal axes are the schools they should optimally have chosen given their preference points and the ex ante admission probabilities. Ex ante admission probabilities are in parentheses. The left-hand graph shows all students who made a mistake due to biased beliefs, the right-hand graph shows all students who made a mistake due to optimization errors. Numbers refer to numbers of students at specific choice/optimal choice combinations. The green circles scale with the number of students in a cell, to guide the eye toward the largest clusters.

Figure 2: Strategic mistakes: choice vs. most-preferred school



Notes: The graphs plot all students who made a strategic mistake. On the vertical axis are the schools they actually chose and on the horizontal axes are the schools they preferred most. Ex ante admission probabilities are in parentheses. The left-hand graph shows all students who made a mistake due to biased beliefs, the right-hand graph shows all students who made a mistake due to optimization errors. Numbers refer to numbers of students at specific choice/most-preferred school combinations. The green circles scale with the number of students in a cell.

Appendix C gives a graphical representation of the classification, in which each student is a point in a plane defined by the relative value and the relative admission probability of the chosen school, and mistakes lie below a hyperbola.

**Why more mistakes than in Amsterdam?** The 18.1% mistake rate is about twice the 8% that De Haan et al. (2023) report for the Amsterdam match of 2013 with a closely related method. Several differences between the two settings can contribute to this gap, and our data do not allow us to say how much each of these differences contributes. First, the Utrecht market is smaller and thinner: with eleven schools and around 1,000 pre-university-track applicants, admission probabilities vary more from year to year, which makes them harder to predict at the time of application. Second, admission probabilities were less transparent in Utrecht. In Amsterdam, running applicant counts per school were published during the application window, whereas Utrecht students had to infer admission probabilities from past years. Third, the two gymnasia in Utrecht also draw applicants from outside the city who are not observed by Utrecht families, adding uncertainty about the competition. The Utrecht setting thus offers more scope for mistakes than the Amsterdam setting.

## 5.2 Who makes mistakes and what they cost

**Who makes which type of mistake?** Table 5 reports sample means of three student characteristics separately for those who did not make a strategic mistake, those whose mistake is due to biased beliefs, and those whose mistake is an optimization error. The three characteristics are the share of female students, the standardized test score, and the share with below-median household income. Columns (4)–(6) report  $p$ -values from pairwise comparisons. Mistake-making is not concentrated in any observable group. The test score and the share of students from below-median-income households do not differ significantly across the three groups. The low-income share is 11% in the no-mistake group, 10% among biased-belief mistake-makers, and 14% among optimization-error makers ( $p$ -value = 0.58 for the largest contrast). The only pairwise contrast below conventional significance is the share of female students. This share is lower among biased-belief mistake-makers than in the no-mistake group (31% against 51%,  $p = 0.03$ ). With six pairwise tests in the table we do not attach much weight to it. In particular, we find no socioeconomic gradient in mistake-making.

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5.4). Figures B2 and B3 in Appendix B present results based on interim admission probabilities. These results are virtually identical.

Table 5: Student characteristics by mistake type

|                         | No mistake<br>(1) | Biased beliefs<br>(2) | Opt. error<br>(3) | (1) vs (2)<br><i>p</i> -value | (1) vs (3)<br><i>p</i> -value | (2) vs (3)<br><i>p</i> -value |
|-------------------------|-------------------|-----------------------|-------------------|-------------------------------|-------------------------------|-------------------------------|
| Female                  | 0.51<br>(0.50)    | 0.31<br>(0.47)        | 0.39<br>(0.49)    | 0.03                          | 0.10                          | 0.49                          |
| Standardized test-score | −0.02<br>(1.00)   | 0.07<br>(1.00)        | −0.01<br>(1.01)   | 0.65                          | 0.94                          | 0.75                          |
| Income below median     | 0.11<br>(0.32)    | 0.10<br>(0.31)        | 0.14<br>(0.35)    | 0.87                          | 0.58                          | 0.61                          |
| # Students              | 353               | 29                    | 49                |                               |                               |                               |

Notes: Standard deviations in parentheses. Columns (4)–(6) report *p*-values from pairwise OLS regressions of the characteristic on a group indicator, with robust standard errors.

**Welfare losses.** Table 6 reports descriptive statistics of the welfare loss for the two groups of mistake-makers. The welfare loss of a mistake-maker is the difference in expected utility, as defined in equation (1), between the optimal choice and the actual choice, evaluated at the actual admission probabilities and the modeled value of the second round. Welfare loss is measured in preference points, the scale on which students rate schools relative to their top school (which receives 100 points). Students who make a mistake due to biased beliefs lose on average 5.56 preference points, with a median loss of 3.45 points. Students who make an optimization error lose much more, on average 18.45 preference points, though their median loss is 5.00 points. The loss distributions are strongly right-skewed: half of all mistake-makers forgo less than 5 points, whereas the 90th-percentile loss among optimization-error students is 69.30 points, more than five times that among biased-beliefs students (12.94 points). The welfare cost of mistakes is thus concentrated: a small group of students with large optimization errors accounts for most of the aggregate loss.

Table 6: Welfare loss by mistake type

|                     | Biased beliefs<br>(1) | Opt. error<br>(2) | All mistakes<br>(3) |
|---------------------|-----------------------|-------------------|---------------------|
| Mean (pref. points) | 5.56                  | 18.45             | 13.66               |
| SD                  | (5.94)                | (26.76)           | (22.33)             |
| p10                 | 0.63                  | 0.70              | 0.67                |
| Median              | 3.45                  | 5.00              | 4.81                |
| p90                 | 12.94                 | 69.30             | 28.30               |
| # Students          | 29                    | 49                | 78                  |

Notes: p10 and p90 denote the 10th and 90th percentiles of the loss distribution; standard deviations in parentheses.

**How much of the optimization-error losses do biased beliefs explain?** Students classified as making an optimization error fail to maximize expected utility under their own beliefs, but their beliefs may be biased as well (Section 4). To quantify how much of their losses biased beliefs can account for, Table 7 decomposes each optimization-error student’s welfare loss into two parts. The belief component is the loss that a perfect optimizer holding the same stated beliefs would also have incurred: the expected-utility gap, under the actual admission probabilities, between the school that is optimal under the actual probabilities and the school that is optimal under the student’s own beliefs. The remainder is the pure-optimization component. The belief component is small: it averages 1.26 of the 18.45 points that optimization-error students lose, and its median is zero. For most of these students, the school that is optimal under their own beliefs coincides with the school that is optimal under the actual probabilities. The two differ for only 6 of the 49 students. Making these students choose optimally at their own, uncorrected beliefs would therefore already recover 17.19 of the 18.45 points. Correcting their beliefs adds at most the remaining 1.26 points, less than a tenth of the loss. The dominance of optimization errors in the welfare accounting is therefore not an artifact of the two categories overlapping.

Table 7: The belief component of optimization-error losses

|                                                            | Total loss<br>(1) | Belief comp.<br>(2) | Pure opt. comp.<br>(3) |
|------------------------------------------------------------|-------------------|---------------------|------------------------|
| Mean (pref. points)                                        | 18.45             | 1.26                | 17.19                  |
| Median                                                     | 5.00              | 0.00                | 4.92                   |
| p90                                                        | 69.30             | 2.06                | 69.30                  |
| Share whose optimal school differs (subjective vs. actual) |                   | 0.12                |                        |
| # Students                                                 |                   | 49                  |                        |

*Notes:* Loss components as defined in the text, evaluated at the actual admission probabilities. The pure-optimization component can be negative when the subjectively optimal school does worse than the actual choice under the actual probabilities. The mean absolute gap between subjective beliefs and actual admission probabilities is 0.23 for students without a mistake ( $N = 293$ ), 0.22 for students with biased beliefs ( $N = 24$ ), and 0.26 for students with an optimization error ( $N = 46$ ).

**What kind of optimization error?** Table 8 divides students who make an optimization error into two distinct types. The larger group, 33 of the 49, are revealed-strategic. These students apply to a school other than their most-preferred one and choose a school that they consider safe (median subjective admission probability 1.00), but they choose the wrong alternative. For 64% of them the chosen school is dominated under their own beliefs, in the sense that another school is both more preferred and at least as safe. The remaining

16 students are sincere: they apply to their most-preferred school even though, according to their own beliefs, their admission probability there is only 0.50 (median). For 75% of them a school that is less preferred but safer would raise their own expected utility. These students take too much risk, like the sincere applicants in [Pathak and Sönmez \(2008\)](#) and unlike the overcautious applicants that [He \(2017\)](#) documents in Beijing. The two types forgo similar amounts under their own beliefs: the mean expected-utility shortfall is 19.9 points for the revealed-strategic group and 17.0 points for the sincere group (19.0 points across all 49). Most optimization errors are thus made by students who try to behave strategically but do not succeed. This underlines why information about admission probabilities has limited value in this setting: these students already act on beliefs about admission probabilities, but they do not act on them correctly. With only 16 students in the sincere group, the comparison between the two types is merely descriptive. For the sincere group a further caveat applies: applying to one’s most-preferred school despite a known admission risk can reflect motives that our preference points do not capture, such as anticipated regret, in which case part of what we label an optimization error is coherent non-expected-utility behavior.

Table 8: The structure of optimization errors

|                         | N<br>(1) | Subjective<br>admission<br>(2) | Too risky<br>(%)<br>(3) | Dominated<br>(%)<br>(4) | EU shortfall<br>(points)<br>(5) |
|-------------------------|----------|--------------------------------|-------------------------|-------------------------|---------------------------------|
| Sincere                 | 16       | 0.50                           | 75                      | 25                      | 17.0                            |
| Revealed strategic      | 33       | 1.00                           | 3                       | 64                      | 19.9                            |
| All optimization errors | 49       | 0.90                           | 27                      | 51                      | 19.0                            |

*Notes:* Too risky: the chosen school has higher preference-point utility but a lower subjective admission probability than the school that maximizes expected utility under the student’s own beliefs. Dominated: another school has both higher utility and at least as high a subjective admission probability. EU shortfall: mean expected-utility difference, in preference points, between the own-belief optimal school and the chosen school.

**Can non-expected-utility behavior explain the optimization errors?** Table 9 examines this further. For each of the 49 optimization errors, it asks whether the actual choice is consistent with three alternatives to expected-utility maximization, all evaluated at the student’s own beliefs. The three alternatives are computed as follows. Maximin asks whether the actual choice maximizes the worst possible outcome, that is, the minimum of the chosen school’s value and the fallback value after a rejection. Minimax regret asks whether the actual choice minimizes the largest possible regret across lottery outcomes, where regret is the difference between the best outcome the student could have secured under that realization and the

outcome obtained. The lottery bonus computes, for each student, the smallest utility bonus on winning the chosen lottery that makes the actual choice optimal under the student’s own beliefs.

A third of the choices maximize the worst-case payoff and a fifth minimize worst-case regret. For 35% of the students, a utility bonus of less than 5 points on winning the chosen lottery would rationalize the choice; the median required bonus is 11.8 points. Probability weighting in the sense of [Prelec \(1998\)](#), which transforms the subjective probabilities before the expected-utility comparison, explains almost nothing (2 of 49). Taken together, 47% of the optimization errors are consistent with at least one of these models.

Two conclusions follow from this. First, part of what we label optimization errors may reflect coherent non-expected-utility behavior, which calls for caution in interpreting the number of mistakes. Second, the welfare conclusion is unaffected: the choices that can be rationalized involve much smaller losses (a median of 2.8 points) than the choices that no model rationalizes (a median of 13.5 points), and the latter account for 73% of the group’s welfare loss. Moreover, in the lottery-bonus model, rationalizing the top decile of losses requires a bonus of more than 100 points on a 100-point scale, which is implausibly large.

Table 9: Non-expected-utility rationalization of the optimization errors

|                                                         | Share<br>(1) | # Students<br>(2) |
|---------------------------------------------------------|--------------|-------------------|
| Maximin optimal                                         | 0.33         | 16                |
| Minimax-regret optimal                                  | 0.20         | 10                |
| Lottery bonus $b < 5$ points                            | 0.35         | 17                |
| Lottery bonus $b < 10$ points                           | 0.45         | 22                |
| Prelec weighting ( $\gamma = 0.65$ )                    | 0.04         | 2                 |
| Prelec weighting ( $\gamma = 0.5$ )                     | 0.04         | 2                 |
| Any of maximin/regret, $b < 5$ , Prelec $\gamma = 0.65$ | 0.47         | 23                |
| Median lottery bonus $b$ (pref. points)                 |              | 11.77             |
| # Students                                              |              | 49                |

*Notes:* All criteria use the student’s stated admission probabilities, elicited preference points, and the round-2 outside option. Applying to a school is a lottery that pays its preference points if admitted and the outside option if rejected. Maximin: the chosen school maximizes the worst-case payoff, taken over the lottery outcomes with positive stated probability. Minimax regret: the chosen school minimizes the worst-case regret over the independent lottery outcomes of all listed schools; the worst-case regret of a school is the best payoff attainable at any other school minus its own worst-case payoff, bounded below by zero. Ties count as rationalized. Lottery bonus:  $b$  is the smallest premium on the chosen school’s preference points that makes it the subjective expected-utility maximizer,  $b = (\max_q EU(q) - EU(c))/a(c)$  with  $c$  the chosen school and  $a(c)$  its stated admission probability. Prelec weighting ([Prelec, 1998](#)): each stated round-1 chance  $a$  is replaced by  $w(a) = \exp(-(-\ln a)^\gamma)$ , the outside option is unchanged, and the student is rationalized if the chosen school then maximizes subjective expected utility under the tie-breaking rule of the baseline classification.

### 5.3 Welfare: Boston versus DA-STB

**Simulation design.** Table 10 compares mean student welfare across assignment mechanisms and mistake-correction scenarios. Because preferences and beliefs are observed only for survey respondents, all scenarios are simulated in the same market of the 431 respondents. To make this respondent market exactly as competitive as the actual 2016 match, we scale each school’s capacity down in proportion to the respondents’ share of all students competing for that school.<sup>14</sup> The students competing for a school include the out-of-district applicants: the 39 and 89 students who apply from outside Utrecht to the two gymnasias, CGU and USG, which do not grant residence priority. With this scaling, the simulated admission probability at each school approximately equals its counterpart in the full market, so the actual admission probabilities used to classify mistakes describe the simulated market as well. We rescale capacities rather than match non-respondents to observably similar respondents, for two reasons. First, matching would require imputing not only preference points but also subjective admission beliefs for students we did not survey, and beliefs are the central object of this paper. Second, the rescaling is school-specific, so it reproduces each school’s competitiveness exactly even when survey response varies across schools, whereas matching on observables does so only as well as the observables predict application behavior. Matching within the school of application would also reproduce competitiveness, but it then amounts to the same within-school reweighting as the rescaling, while still requiring the imputed beliefs.

**Welfare measure and scenarios.** Welfare in each scenario is computed over 1,000 simulation draws. In each draw, round-1 seats are allocated by sibling priority and a single lottery, and students left unplaced are assigned through a single-shot round-2 Boston match over the residual seats, in which each applies to their most-preferred remaining school and rejected applicants are placed in their best still-open school. The second round is thus re-simulated in every draw, since which seats remain depends on that draw’s lottery realization. A student’s welfare is the mean preference-point value of her assigned school across draws. Averaging preference points across students requires an interpersonal comparison of utility, for which Segal (2000) provides a foundation.<sup>15</sup> Section 5.4 repeats the comparison under criteria that

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<sup>14</sup>Table B1 in Appendix B reports, for each school, the number of first-round applicants and the full and survey capacities.

<sup>15</sup>A weighted sum of von Neumann–Morgenstern utilities is in general sensitive to each individual’s arbitrary cardinal normalization. Under the axiom that society considers assigning the entire endowment to any single individual equally bad, the welfare-relevant weights equalize each individual’s maximum attainable

do not use a cardinal scale (Table 13).

The scenarios we compare are (i) actual Boston, in which each student submits their real first-round choice; (ii) corrected-beliefs Boston, in which the 29 biased-belief mistake-makers play optimally given the actual admission probabilities, while all other students keep their actual choices; (iii) optimal-choices Boston, in which all students play optimally; and (iv) DA-STB, in which each student submits their full stated preference ranking, which under DA is optimal play, and all schools break ties by the same single lottery draw per student (Section 3.2), as in the mechanism comparison of Abdulkadiroğlu et al. (2011). In scenarios (ii) and (iii) we solve for the Nash equilibrium of the first-round application game, in which the students who re-optimize, the 29 biased-belief mistake-makers in scenario (ii) and all respondents in scenario (iii), best-respond to the admission probabilities that the joint choices imply. The solution method is described below. Panel A of the table reports welfare levels, Panel B reports within-student changes relative to actual Boston, and Panel C reports the direct contrast between optimal-choices Boston and DA-STB.

**Computing the equilibrium.** We solve the equilibrium scenarios by best-response iteration: starting from a profile of first-round choices, we recompute the admission probabilities from the applicant pool that the profile implies, let every student best-respond to these probabilities, and repeat until the profile reproduces itself. The iteration converges within a few steps. Because best-response dynamics in this game could in principle have multiple fixed points, we start the iteration from different profiles, including the actual choices and the choices that are optimal under the baseline admission probabilities; it reaches the same fixed point from each of them. With only two schools oversubscribed in this market, the scope for multiplicity is limited. The equilibrium scenario assumes that every respondent best-responds with correct beliefs. It is thus the benchmark of fully sophisticated play.

**Aggregate welfare.** We discuss the table column by column, starting with all students together (column (1)). Under actual Boston, mean welfare across all students is 93.46 preference points, against 94.67 under DA-STB. DA-STB thus gives 1.21 points more than Boston. When all students play optimally, Boston yields 94.46 points in equilibrium. Removing the mistakes thus closes most of the gap to DA-STB but does not produce an advantage. The ag-

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utility. Our points scale implements this by setting every student's most-preferred school equal to 100, so that averaging preference points across students is well defined and invariant to the underlying cardinal representation. This normalization is nonetheless a substantive ethical choice, and the mechanism comparisons below involve welfare gaps of one to two points.

Table 10: Mean welfare by mechanism and mistake type

|                                                  | All<br>(1)        | No mistake<br>(2)  | Biased beliefs<br>(3) | Opt. error<br>(4)   |
|--------------------------------------------------|-------------------|--------------------|-----------------------|---------------------|
| <i>Panel A: Mean welfare (preference points)</i> |                   |                    |                       |                     |
| Actual Boston                                    | 93.46<br>(0.66)   | 96.55<br>(0.33)    | 85.76<br>(1.86)       | 75.78<br>(4.28)     |
| Corrected beliefs (equilibrium)                  | 93.20<br>(0.64)   | 95.73<br>(0.35)    | 88.04<br>(1.92)       | 77.99<br>(4.25)     |
| Optimal choices (equilibrium)                    | 94.46<br>(0.45)   | 95.40<br>(0.44)    | 86.12<br>(2.49)       | 92.66<br>(1.49)     |
| DA-STB                                           | 94.67<br>(0.37)   | 95.13<br>(0.39)    | 91.64<br>(1.19)       | 93.18<br>(1.45)     |
| Constrained-efficient (ceiling)                  | 99.03<br>(0.13)   | 99.13<br>(0.14)    | 98.72<br>(0.49)       | 98.55<br>(0.42)     |
| <i>Panel B: Change from actual Boston</i>        |                   |                    |                       |                     |
| Corrected beliefs (equilibrium)                  | -0.27<br>(0.26)   | -0.82***<br>(0.30) | +2.28***<br>(0.68)    | +2.21***<br>(0.53)  |
| Optimal choices (equilibrium)                    | +1.00<br>(0.64)   | -1.15***<br>(0.41) | +0.36<br>(1.26)       | +16.87***<br>(4.12) |
| DA-STB                                           | +1.21**<br>(0.55) | -1.42***<br>(0.28) | +5.88***<br>(0.94)    | +17.40***<br>(3.55) |
| <i>Panel C: Difference from DA-STB</i>           |                   |                    |                       |                     |
| Optimal choices (equilibrium) – DA-STB           | -0.21<br>(0.29)   | +0.27<br>(0.29)    | -5.52***<br>(1.72)    | -0.52<br>(1.05)     |
| # Students                                       | 431               | 353                | 29                    | 49                  |

*Notes:* The table compares mean student welfare across four scenarios, simulated for the 431 survey respondents with school capacities scaled to the survey sample (Table B1), sibling priority, and a single lottery number per student per draw (1,000 draws). Scenarios as defined in the text: in the Nash equilibrium of the application game, the 29 biased-belief mistake-makers play optimally while all other students keep their actual choices (corrected beliefs), or all students play optimally (optimal choices); under DA-STB each student submits their full stated ranking. Welfare is the mean preference-point score of the assigned school across draws (0–100 scale; top school = 100). The constrained-efficient ceiling maximizes total preference points subject to capacities (full cardinal information, no priorities or lotteries). Panel C reports the within-student difference between the optimal-choices Boston counterfactual and DA-STB. Standard errors, in parentheses, are cross-student sampling errors, not simulation errors: in Panel A they are computed across students within each column, and in Panels B and C from the within-student (paired) differences. Stars denote significance from a within-student  $t$ -test: \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

gregate gain from removing all mistakes (1.00 points) is of the same order as the advantage of DA-STB over actual Boston, while the individual losses of the mistake-makers themselves are much larger (Table 6). Panel C reports the direct within-student test of this key contrast. Optimal-choices Boston minus DA-STB is  $-0.21$  points (s.e. 0.29,  $p = 0.47$ ) in equilibrium. Abdulkadiroğlu et al. (2011) show that Boston can give higher welfare than DA-STB when students play optimally. In this market, the two mechanisms are statistically indistinguishable under optimal play, with point estimates slightly favoring DA-STB. The equilibrium confidence interval rules out a Boston advantage larger than 0.4 points, which is a third of the advantage that DA-STB holds over actual play. Correcting only biased beliefs leaves aggregate welfare essentially unchanged ( $-0.27$  points in equilibrium, not significant), because most of the welfare loss under Boston is due to optimization errors rather than biased beliefs.

To put these magnitudes in perspective, even the constrained-efficient assignment reaches 99.03 preference points rather than 100, because several schools are oversubscribed. The constrained-efficient assignment allocates the 431 respondents to seats so as to maximize mean preference points subject to the rescaled capacities, ignoring incentives and lotteries. We compute it as a linear assignment problem, and it serves only as an upper benchmark for attainable welfare. The maximum attainable welfare gain relative to students' actual choices is therefore 5.57 points. Switching to DA-STB realizes 22% of this gain, and letting every student play Boston optimally realizes 18% in equilibrium. Removing mistakes thus raises aggregate welfare by less than the individual losses in Table 6 suggest. The mean loss of 13.66 points among 78 mistake-makers corresponds to 2.47 points per respondent. The reason is that seats are scarce: part of what a mistake-maker forgoes is captured by the student who obtains the seat instead, so correcting mistakes partly redistributes welfare rather than creating it.

**Who gains and who loses.** The remaining columns of Table 10 break the comparison down by mistake type. Students who made no mistake (column (2)) prefer the status quo. They lose 1.42 points under DA-STB, because DA takes away the advantage that their correct strategic play earns under Boston. The correction scenarios also cost them a little, between  $-0.8$  and  $-1.2$  points, because corrected rivals compete more effectively for the same seats.

Biased-belief mistake-makers (column (3)) gain substantially from DA-STB (+5.88 points). They gain much less from having their beliefs corrected under the Boston mechanism: correcting beliefs raises their welfare by 2.28 points. This is well short of the +5.88 points they obtain under DA-STB, where a student who misjudges admission probabilities is no longer penalized for ranking a competitive school first. What helps biased-belief students most is

the strategy-proof mechanism, not information provision while the manipulable mechanism stays in place.

Optimization-error students (column (4)) face by far the largest welfare losses (Table 6). They gain the most from any remedy. Switching them to optimal choices raises their mean welfare by 16.87 points. DA-STB gives them slightly more still (+17.40 points). Correcting beliefs raises their welfare by 2.21 points even though their own choices do not change. The corrected choices of biased-belief students reduce competition at some schools, and optimization-error students benefit through this channel.

**Why optimal-choices Boston does not outperform DA in this market.** [Abdulkadiroğlu et al. \(2011\)](#) and [Troyan \(2012\)](#) show that the Boston mechanism can deliver higher welfare than the DA mechanism. The channel is sorting by preference intensity. Under Boston, applying to an oversubscribed school means giving up the first-round claim to a safe school. When the value of the safe school relative to the fallback is the same for all students, the students who apply to the oversubscribed school are those who gain most from a seat relative to the fallback they face after a rejection, and the scarce seats go to these students. Under DA, the seats are allocated by priority and lottery among all students who rank the school highly, regardless of how much they gain from a seat.<sup>16</sup>

Our simulations show no such advantage for Boston. This does not contradict the theory, because the theory guarantees a Boston advantage only from a particular welfare perspective. [Troyan \(2012\)](#) shows that Boston dominates DA ex ante, that is, before students know their own cardinal utilities and priorities. Once students know their own utilities and priorities, neither mechanism needs to dominate the other.<sup>17</sup> Our simulations evaluate welfare at students' realized utilities and priorities, where no dominance result applies. Whether Boston does better then depends on whether the sorting channel operates in the market at hand. In Utrecht it does not operate, for two reasons. A third feature of the market works against Boston directly.

First, for many applicants to the oversubscribed schools, applying to a safer school yields little. What a safer application adds is the value of the safer school above the round-2 fallback. This depends on whether the safer school still has seats in the second round. Of the 105

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<sup>16</sup>Both papers derive this result in an environment where students share the same ordinal ranking over schools. In that environment every student gives up the first-round claim to the same safe school and faces the same fallback. Utilities can then be normalized so that these two schools have the same values for every student, and only the value of the oversubscribed school varies across students.

<sup>17</sup>The terminology differs across papers: what [Abdulkadiroğlu et al. \(2011\)](#) and we call ex-ante welfare corresponds to Troyan's interim concept.

students who rank CGU first, 39 have a best alternative that keeps seats open in the second round. A rejected student can then often still obtain this school, or one close to it, in round 2. For these students the best alternative is worth a median of only 5.0 preference points more than the fallback, and 85% of them optimally apply to CGU. At Gerrit Rietveld College the corresponding median is zero, and 93% optimally apply there. When the safer application yields almost nothing extra, the student applies to the oversubscribed school whatever her gain from a seat, and the mechanism cannot sort students on that gain. A modification of the example in [Abdulkadiroğlu et al. \(2011, p. 402\)](#) makes the point. In their example three students compete for three schools with one seat each and share the ordinal ranking  $s_1 \succ s_2 \succ s_3$ . Students 1 and 2 have utilities  $(0.8, 0.2, 0)$  over the three schools and student 3 has  $(0.6, 0.4, 0)$ . Every student values the fallback school  $s_3$  at zero. In the unique equilibrium of the Boston mechanism, student 3, whose preference for  $s_1$  is least intense, applies to  $s_2$  and is admitted with certainty, while students 1 and 2 apply to  $s_1$ . Every student obtains an expected utility of 0.4, against  $1/3$  under DA. Now raise student 3's value of the fallback school from 0 to 0.3. Applying to  $s_2$  then yields little more than applying to  $s_1$ , and student 3's best response is to apply to  $s_1$  as well (an expected utility of 0.43 against 0.40). In the resulting equilibrium all three students apply to  $s_1$ , the assignment coincides with the DA lottery, and the Boston advantage disappears.

Second, when some students do apply to a safer school, fallbacks that differ across students mean they need not be those with the smallest gain from a seat.<sup>18</sup> At CGU, the students who optimally apply elsewhere gain more from a seat relative to their fallback (a mean of 55.6 points) than the students who keep applying (43.7 points). The same pattern holds at Gerrit Rietveld College (50.2 against 34.6 points). The pool of applicants under optimal Boston therefore does not overrepresent the students with the largest welfare gain from a seat.

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<sup>18</sup>The following example illustrates this. Safe schools admit all applicants in the first round but are full afterwards, so a student who applies to the oversubscribed school is admitted with probability  $p$  and otherwise ends up at her fallback. She applies to the oversubscribed school if and only if  $p(v_{\text{over}} - v_{\text{fall}}) > v_{\text{safe}} - v_{\text{fall}}$ , that is, if and only if her gain from a seat exceeds the threshold  $(v_{\text{safe}} - v_{\text{fall}})/p$ . When the safe-school and fallback values are the same for all students, this threshold is common and the applicants are exactly the students with the largest gains. When these values are student-specific, the threshold is student-specific and the sorting breaks down. Anna and Ben both award an oversubscribed school 100 points; its admission probability is 0.7, determined by a larger pool of applicants. Anna's safe school is worth 95 points to her and her fallback 60 points; for Ben these values are 81 and 80 points. Anna's threshold is  $(95 - 60)/0.7 = 50$  points, so with a gain of 40 points she applies to her safe school. Ben's threshold is  $(81 - 80)/0.7 \approx 1.4$  points, so with a gain of 20 points he applies to the oversubscribed school. Boston thus assigns the seat to the applicant with the smaller welfare gain. A lottery among both, as under DA, would in expectation allocate it better (a mean gain of 30 points).

Third, immediate acceptance is costly for rejected students: a student who is rejected can no longer apply to schools that filled their seats in the first round. This cost is present in every comparison of Boston with DA, but it is large in Utrecht. USG and St. Bonifatius College, the natural second choices of CGU applicants, fill their seats in round 1. A student who applies to CGU and is rejected therefore loses access to her second choice. This is the counterpart of the first feature. Of the 105 students who rank CGU first, 60 have a best alternative that fills its seats in round 1, for 38 of them USG and for 13 St. Bonifatius College. For these students the alternative is worth a median of 33.0 preference points more than the round-2 fallback, and this is what a rejection at CGU costs them. Under DA-STB, deferral preserves this access: a rejected CGU applicant competes for USG on an equal footing with everyone else.

Combining the small and possibly absent sorting gain with this cost yields our estimate of  $-0.21$  points, which does not differ significantly from zero. The role of the fallback also explains why the sign of the small optimal-play difference varies with the modeled value of the second round in Section 5.4: the value of the second round determines both the cost of applying to the oversubscribed school and the sorting it induces.

**Sorting by household income.** Beyond mean welfare, the mechanism could shape how students sort across schools by household income. Table B2 reports, for each allocation, the Duncan dissimilarity index of low-income against other students across schools and the share of each group placed in one of the two single-track gymnasia. We single out the two gymnasia because they are the schools that would typically be selective on ability in comparable settings, so access to them is where a change of mechanism could most plausibly change low-income students' outcomes. Under students' actual choices the dissimilarity index is 0.17, and it is similar under the counterfactuals and under DA-STB (0.13). These differences are too small to be distinguished from sampling noise: with only 50 low-income students, no allocation's index differs significantly from actual Boston, and the gap between actual Boston and DA-STB (0.04) has a  $p$ -value of 0.45. Access to the gymnasia is the same for low-income and other students under every allocation, at about one in four, with no significant difference. The choice of mechanism thus has no detectable effect on the distribution of low-income students across schools. This contrasts with settings in which schools select on ability, where under the manipulable mechanism sophisticated families apply less often to selective schools, which can raise disadvantaged students' access to these schools (Terrier et al., 2026). The Utrecht gymnasia admit by lottery rather than ability, so that channel does not operate here.

## 5.4 Robustness

**Imputation of missing beliefs.** Table 11 probes the treatment of missing subjective admission beliefs systematically. It recomputes the classification and the welfare losses under four rules: the certain-admission imputation of the main text (missing = 100%), the opposite extreme (missing = 0%), imputation at the school’s actual admission probability, and dropping the 49 respondents who report no belief for any school. Three results stand out. First, the total number of strategic mistakes is 78 under every imputation rule. Beliefs enter only the classification step, not the identification of mistakes. Second, the classification shifts with the rule, as it must, but optimization errors remain the majority in every scenario. The biased-belief count ranges from 24 (actual-probability imputation) to 31 (missing = 0%), against 47 to 54 optimization errors. Third, the welfare-loss statistics barely move. The mean loss across all mistakes is 13.66 points under every imputation rule and 13.10 when the 49 non-reporters are dropped. Dropping them yields 72 mistakes among the remaining 382 respondents (18.8%), of which 24 are due to biased beliefs and 48 to optimization errors. The certain-admission imputation does push the 49 non-reporters themselves toward the biased-beliefs category (five of their six mistakes), but they are too few to move the main-text split.

Table 11: Robustness to the imputation of missing beliefs

|                              | Resp.<br>(N)<br>(1) | Mistakes<br>(N) (%)<br>(2) (3) | Beliefs<br>(N)<br>(4) | Opt. err.<br>(N)<br>(5) | Mean welfare loss |                |                  |
|------------------------------|---------------------|--------------------------------|-----------------------|-------------------------|-------------------|----------------|------------------|
|                              |                     |                                |                       |                         | All<br>(6)        | Beliefs<br>(7) | Opt. err.<br>(8) |
| Baseline: missing = 100%     | 431                 | 78 18.1                        | 29                    | 49                      | 13.66             | 5.56           | 18.45            |
| Missing = 0%                 | 431                 | 78 18.1                        | 31                    | 47                      | 13.66             | 6.33           | 18.50            |
| Missing = actual probability | 431                 | 78 18.1                        | 24                    | 54                      | 13.66             | 5.93           | 17.10            |
| Drop no-belief respondents   | 382                 | 72 18.8                        | 24                    | 48                      | 13.10             | 5.63           | 16.83            |

*Notes:* Imputation rules as described in the text, applied to school-beliefs left blank. Column (1): respondents; columns (2)–(3): strategic mistakes (number, % of respondents); columns (4)–(5): mistakes attributed to biased beliefs and to optimization errors; columns (6)–(8): mean welfare loss (expected-utility difference between the optimal and the chosen school, preference points) over mistake-makers of the given type.

**Estimated instead of elicited cardinal utilities.** Appendix D derives results using cardinal utilities estimated from a rank-ordered logit on students’ stated school rankings, instead of the survey-elicited preference points. Fewer strategic mistakes are detected (11.4% versus 18.1%), but the qualitative conclusions are unchanged: optimization errors dominate, neither measure delivers a Boston advantage over DA-STB under optimal play, and correcting beliefs alone has a negligible aggregate effect. Table 12 shows that the welfare decomposi-

tion is nearly identical under the two measures. Optimization errors account for 84.9% of the total welfare loss under the elicited points and for 90.5% under the estimated utilities.

Table 12: The welfare decomposition under the two preference measures

|                                               | Elicited points<br>(1) | Estimated utilities<br>(2) |
|-----------------------------------------------|------------------------|----------------------------|
| # Strategic mistakes                          | 78                     | 49                         |
| # Optimization errors                         | 49                     | 42                         |
| Optimization errors (% of mistakes)           | 62.8                   | 85.7                       |
| Optimization errors (% of total welfare loss) | 84.9                   | 90.5                       |
| Mean loss, biased-belief students             | 5.56                   | 31.65                      |
| Mean loss, optimization-error students        | 18.45                  | 50.03                      |

*Notes:* Column (1) measures utility in elicited preference points (0–100); column (2) in utilities estimated from a rank-ordered logit (Appendix D). The two scales differ, so the mean losses in the last two rows are not comparable across columns; the shares are unit-free.

Between the two measures, we regard the elicited preference points as the more credible welfare metric. The points carry cardinal information by design: the elicitation defines them as probability equivalents relative to the top school, so they measure intensity directly. The rank-ordered logit sees only the ordinal ranking, and the cardinal spread it assigns to the ranks comes from the logistic functional form rather than from data. Appendix D shows that this feature, a steeper imputed utility drop below the top rank, accounts for nearly all of the difference in mistake counts.

**Sensitivity of the classification.** Appendix E probes the classification along three further dimensions. First, “optimization error” is a residual category, so noise in the elicited preference points could create spurious mistakes and assign them to that category. Noise can make a correct choice look suboptimal: when measurement error understates the value of the chosen school relative to an alternative, the classification records a mistake the student did not make. Because the stated beliefs are elicited separately and are unaffected by this noise, such a spurious mistake can typically not be rationalized by the beliefs, so it is assigned to the residual optimization-error category. The simulation adds mean-zero normal noise to the elicited points of every ranked school below the top rank, rebuilds the round-2 outside option from the noisy points, and reruns the full classification 200 times per noise level. The effect is present but small. At a noise standard deviation of 5 points the mean mistake count is 78.7, against 78 at baseline. The optimization-error majority, in counts and in welfare loss, survives in every one of 200 draws, even at a standard deviation of 10 points, which is about half of the mean gap between first- and second-ranked schools. Second, we

perturb the modeled value of the second round: direct shifts of  $\pm 25\%$  and  $\pm 10$  points, shifted round-2 admission probabilities, better or worse fallback rules, and removing or discounting the unrationed outside-Utrecht alternative. The mistake count moves between 65 and 98, but optimization errors remain the majority (58–74% of mistakes) under every variant. The small optimal-play welfare difference between Boston and DA-STB does change sign within this band. Third, about a quarter of the mistake-makers are within 10% of the classification boundary, while the mistakes with the largest losses are far from the boundary. The belief/optimization split itself is robust: only 21% of biased-belief mistakes and 11% of optimization-error mistakes are within 10% of their boundary.

**Alternative welfare criteria.** The welfare comparison of Section 5.3 averages preference points across students. This is one of several defensible welfare criteria, and the comparison should not depend on it. Table 13 therefore repeats the comparison under five alternatives, computed from the same simulations. The median welfare level describes the typical student. The bottom-quartile and bottom-decile means describe the students for whom the choice of mechanism matters most. The mean assigned rank and the share of students assigned to their top choice use only the stated ordinal ranking and no cardinal information. Panel B reports within-student comparisons with DA-STB. The two panels select different students: the bottom-quartile mean in Panel A averages each scenario’s own worst-off students, whereas the bottom-quartile difference in Panel B follows the students in the bottom quartile of the DA-STB distribution. The two can therefore diverge when the mechanisms hurt different students. The median paired difference and its sign test show whether more students are better or worse off than under DA-STB, again without using the cardinal scale.

The criteria do not all point in the same direction. The median student does slightly better under Boston. The median welfare level is higher under actual and optimal Boston than under DA-STB, and among students whose outcome differs under optimal play, 65% prefer the Boston outcome (sign test,  $p < 0.001$ ). The same paired test for actual Boston is a tie (Panel B).

All other criteria favor DA-STB or show no difference. The mean assigned rank is 1.30 under DA-STB against 1.37 under equilibrium optimal Boston ( $p < 0.001$ ), while the share of students assigned to their top choice is the same under both. This rank advantage of DA-STB does not depend on optimal play. Actual Boston and corrected-beliefs Boston place students at a worse mean rank still, 1.39 and 1.40, and Panel B confirms that both within-student gaps are significant (columns (1) and (2),  $p < 0.001$ ).

Correcting beliefs is also the only scenario that lowers the top-choice share: corrected-

beliefs Boston places 3.9 percentage points fewer students in their first choice than DA-STB ( $p < 0.01$ ), because the corrected students redirect to the same safe schools. The advantage of DA-STB is largest in the left tail: the mean welfare of the bottom decile is 76.5 points under DA-STB, against 71.0 under equilibrium optimal Boston and 63.6 under actual Boston. The Boston mechanism concentrates losses on students who apply to an oversubscribed school and are rejected, whereas DA-STB protects these students. The criteria that use no cardinal information, ranks and top-choice shares, agree with the mean-points comparison.

Table 13: The mechanism comparison under alternative welfare criteria

|                                                                          | Actual<br>Boston<br>(1) | Corrected<br>beliefs eq.<br>(2) | Optimal<br>choices eq.<br>(3) | DA-STB<br>(4) | Efficient<br>ceiling<br>(5) |
|--------------------------------------------------------------------------|-------------------------|---------------------------------|-------------------------------|---------------|-----------------------------|
| <i>Panel A: Welfare criteria (per-student mean outcome across draws)</i> |                         |                                 |                               |               |                             |
| Mean (preference points)                                                 | 93.46                   | 93.20                           | 94.46                         | 94.67         | 99.03                       |
| Median                                                                   | 99.12                   | 97.84                           | 99.00                         | 98.29         | 100.00                      |
| Bottom-quartile mean                                                     | 77.87                   | 78.57                           | 81.32                         | 83.85         | 96.15                       |
| Bottom-decile mean                                                       | 63.56                   | 64.90                           | 71.04                         | 76.54         | 91.59                       |
| Mean assigned rank                                                       | 1.39                    | 1.40                            | 1.37                          | 1.30          | –                           |
| Share assigned top choice                                                | 0.793                   | 0.773                           | 0.806                         | 0.812         | –                           |
| <i>Panel B: Scenario – DA-STB (within student)</i>                       |                         |                                 |                               |               |                             |
| Median difference (points)                                               | 0.00<br>[p = 0.495]     | 0.00<br>[p = 0.034]             | 0.00<br>[p < 0.001]           |               |                             |
| Bottom-quartile difference                                               | +0.47<br>(1.60)         | -0.99<br>(1.44)                 | -0.06<br>(1.07)               |               |                             |
| Mean assigned rank                                                       | +0.09***<br>(0.02)      | +0.10***<br>(0.02)              | +0.07***<br>(0.02)            |               |                             |
| Share assigned top choice                                                | -0.019<br>(0.013)       | -0.039***<br>(0.014)            | -0.006<br>(0.011)             |               |                             |
| # Students                                                               | 431                     | 431                             | 431                           | 431           | 431                         |

*Notes:* Alternative welfare criteria for the mechanism comparison of Table 10, computed from the same simulations (431 survey respondents, survey-scaled capacities, 1,000 draws per scenario). Each student is summarized by their mean outcome across draws; the table reports statistics of this per-student distribution. The mean row reproduces Panel A, column (1), of Table 10. Bottom-quartile (bottom-decile) means average the 108 (44) students with the lowest mean utility in each scenario. The assigned rank is the position of the assigned school in the student’s stated ranking; an assignment outside the stated list is scored one rank beyond the list length. Rank-based rows are unavailable for the constrained-efficient ceiling, which is solved as an assignment problem without simulated allocations. Panel B reports within-student differences between each Boston scenario (columns (1)-(3)) and DA-STB; column (4) is DA-STB itself and column (5) has no simulated allocation. The median row shows the median paired difference with the  $p$ -value of a sign test (zero differences dropped) in brackets. The bottom-quartile row averages the paired differences over the 108 students in the bottom quartile of the DA-STB distribution. Standard errors of the paired mean differences are in parentheses; stars denote within-student  $t$ -tests: \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 6 Conclusion

We have used a unique combination of survey and register data from the secondary-school match in Utrecht to distinguish two sources of strategic mistakes: biased beliefs about admission probabilities and optimization errors. Earlier work identifies strategic mistakes as the reason the Boston mechanism can perform worse than DA, but it does not separate these two sources. Our main finding is that 18.1% of the students make a strategic mistake, and that optimization errors account for about two thirds of them, against one third from biased beliefs. Optimization errors are also far more costly: they account for 85% of the total welfare loss from strategic mistakes. Under students' actual choices, mean welfare is lower under Boston than under DA-STB, where truthful ranking is optimal play. Removing all mistakes closes most of this gap but does not reverse it: under optimal play, mean welfare under Boston is statistically indistinguishable from that under DA-STB. The potential theoretical advantage of the manipulable mechanism thus does not materialize in this market. Correcting only biased beliefs has almost no aggregate effect, because optimization errors account for most of the welfare loss.

Our evidence comes from one market. Yet we expect our findings on strategic mistakes to hold more generally. That a sizable share of students make strategic mistakes under Boston, and that most of these mistakes are optimization errors rather than biased beliefs, reflects the difficulty of the choice problem that Boston poses. Students must weigh how much they prefer a school against their admission chances and the value of their fallback. They face this trade-off in any market that uses the mechanism.

These results bear on what a school district running a Boston-type mechanism should do. Providing accurate information about admission probabilities is the natural response to biased beliefs, but it addresses the smaller of the two sources of mistakes and has only a modest effect on welfare. It does little even for the students whose beliefs are corrected, who instead gain substantially from a switch to the strategy-proof mechanism. Information provision and a change of mechanism are therefore not substitutes. The case for DA is strong in this setting: under actual behavior DA delivers higher mean welfare, and under fully optimal behavior Boston delivers no more than DA, so a switch to DA involves no welfare loss. We find no socioeconomic gradient in the mistakes: students who make a mistake do not differ significantly from other students in income or test scores. The mechanism also leaves the distribution of low-income students across schools, including the selective gymnasia, unchanged. The value of information depends on the mechanism. [Larroucau et al. \(2026\)](#) show that personalized information reduces application mistakes in Chile's college admissions.

Under a manipulable mechanism, information can likely fix only part of the mistakes, because most of them are optimization errors. Replacing the mechanism with a strategy-proof alternative may therefore be more effective.

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## A Choice behavior in the second round

Students who were not placed in the first round can submit a new ranked list in a second application round. This appendix describes who participates in the second round and how they behave. Schools are recoded into the same twelve-school analysis set used in the main text, so schools outside the core set are pooled into the “Outside Utrecht” category.

### A.1 Participation and choices

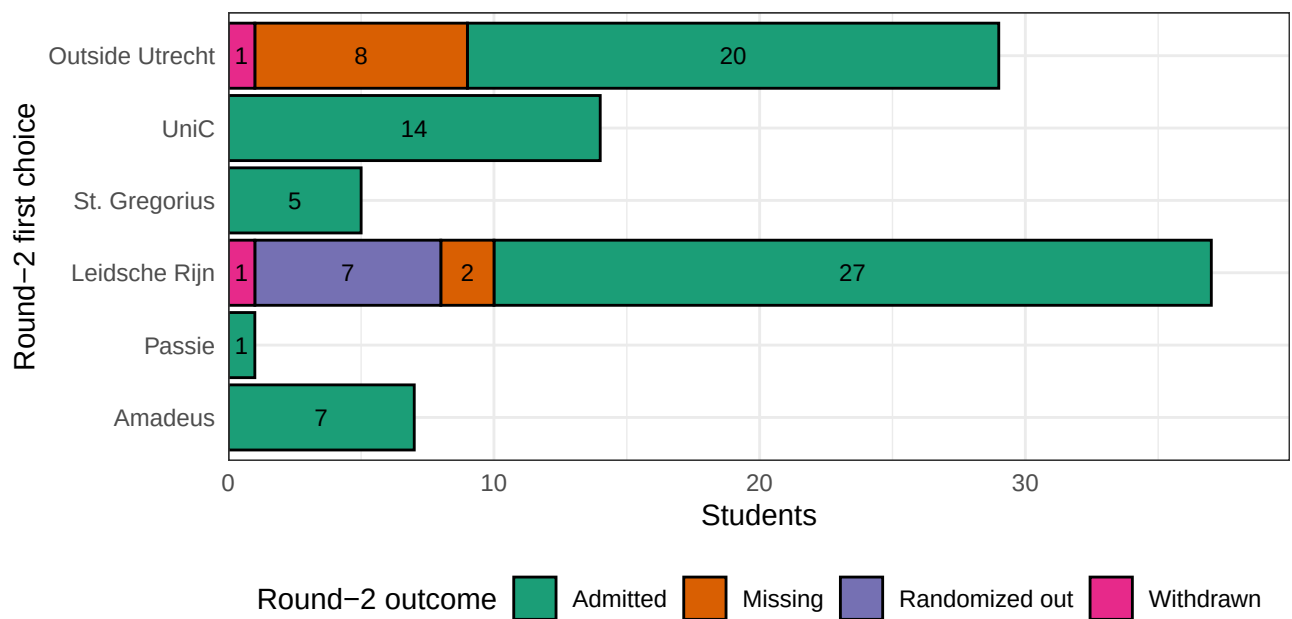
The second round is, in practice, the re-application stage for the students who lost a first-round lottery. Of the 93 students who submit a second-round list, 90 were randomized out of their first-round first choice; the remaining three had been admitted or rejected for other reasons. In the other direction, 90 of the 93 students who were randomized out in the first round participate in the second round, so participation is essentially universal among first-round lottery losers. Almost every participant changes their first choice between rounds: only two of the 93 list the same first choice as in round 1, which is unsurprising given that the round-1 first choice is precisely the school at which they lost the lottery.

**How many schools do students list?** The median student lists two schools and the mean list length is 2.0. Twenty-five of the 93 participants (27%) list only a single school, while the remaining 73% list at least two; ten list three schools and nine list the maximum of four. Second-round lists are therefore short, but most students do leave themselves a fallback.

**What do second-round preferences look like (Figure A1).** Second-round demand is concentrated. Leidsche Rijn College is by far the most popular first choice (37 students), followed by UniC (14) and a long tail of less popular schools; many of the listed schools fall outside the core eleven and are pooled into “Outside Utrecht”. Figure A2 shows the joint distribution of first and second choices, which displays the same concentration: when students do list a second school it is most often Amadeus Lyceum or another non-oversubscribed school, used as a safety option below a popular first choice.

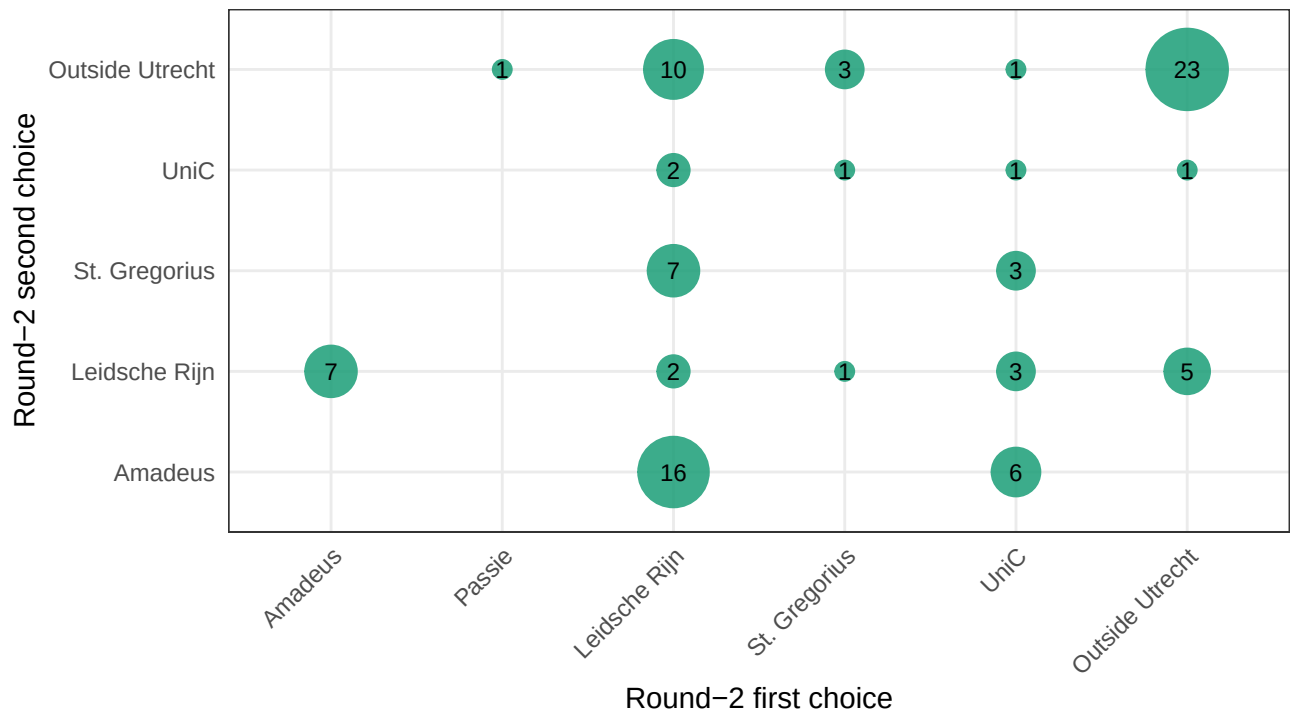
**Second-round preferences conditional on the first round (Figure A3).** Because second-round participants are almost all first-round lottery losers, their second-round first choice is naturally a substitute for the school they failed to enter. The figure groups participants

Figure A1: Second-round first choices, by outcome



*Notes:* The figure shows the schools that second-round participants rank first, with the bar shading indicating the admission outcome on that first choice. “Missing” denotes participants for whom no first-choice decision is recorded. Schools outside the core eleven are pooled into “Outside Utrecht”.

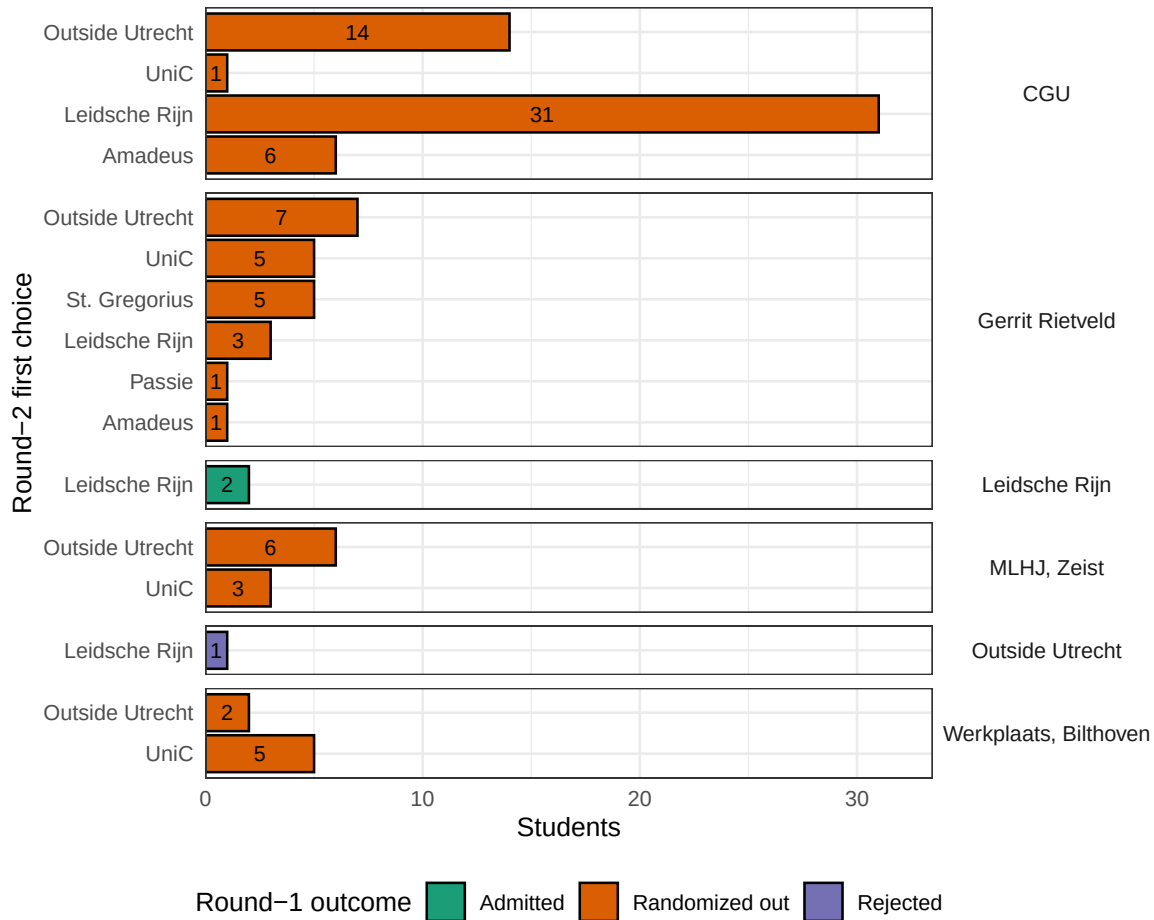
Figure A2: Joint distribution of second-round first and second choices



Notes: The figure plots second-round first choices (horizontal axis) against second choices (vertical axis) for the participants who list at least two schools. The size of each point and the number printed in it give the number of students at that first/second-choice combination.

by their first-round first choice. Students who were randomized out of the two most over-subscribed schools (CGU, Gerrit Rietveld College) reapply mainly to Leidsche Rijn College and to schools outside the core set, while students from other first-round schools spread across UniC, Amadeus Lyceum and Outside Utrecht. The pattern is consistent with students moving down a stable preference order rather than re-optimizing from scratch.

Figure A3: Second-round first choice, by first-round first choice and outcome



*Notes:* The figure shows second-round first choices (horizontal facets) split by the school the student ranked first in the first round (right-hand labels). Bar shading indicates the first-round admission outcome. Almost all participants were randomized out in the first round.

**Alignment of second-round choices with stated preferences.** Of the 93 second-round participants, 47 (50.5%) are respondents to the choice survey used in the main analysis, close to the overall survey response rate. For these 47 participants we can compare their second-

round first choice with the preferences they stated in the survey. None of them re-lists their stated favorite school as their second-round first choice. This is expected, because the stated favorite essentially coincides with the round-1 first choice at which they were rejected (it does so for 46 of the 47). Instead, students fall back down their own stated preference order, with 13 picking their stated second preference and 15 their stated third preference as their second-round first choice. For 12 respondents the second-round first choice does not appear in the stated list at all. On the whole, second-round first choices are well aligned with stated preferences: 35 of 47 (74%) choose a school they had ranked in the survey, and the typical student moves one or two steps down a stable preference ordering after losing the first-round lottery.

## A.2 The value of the second round

This subsection constructs the value of the second round,  $v_i^{r2}$ , which serves as the outside option in the strategic-mistake classification of Section 4. We model the second round as a one-shot Boston stage. A student who reaches round 2 applies to a single school  $s$  and is admitted there with the actual second-round admission probability  $\alpha_s^{r2}$  (Table A1); with probability  $1 - \alpha_s^{r2}$  the student is randomized out and is then placed, without further competition, in the best safe school they ranked, a school with a second-round admission probability of one. By Table A1 the safe schools are St. Gregorius, UniC, and the “Outside Utrecht” fallback, whereas Leidsche Rijn College, Amadeus Lyceum, and De Passie are rationed. Writing  $v_s$  for the cardinal utility a student attaches to a ranked school (the preference points of Section 3, with  $v_s = 0$  for any school they did not rank), the expected utility of applying to  $s$  in round 2 is

$$\text{EU}(s) = \alpha_s^{r2} v_s + (1 - \alpha_s^{r2}) v_{\text{fb}}, \quad v_{\text{fb}} = \max_{t: \alpha_t^{r2}=1} v_t, \quad (\text{A1})$$

where the maximum defining the safe fallback  $v_{\text{fb}}$  is taken over the safe schools the student ranked ( $v_{\text{fb}} = 0$  if they ranked none). The *optimal round-2 choice* is the school  $s^{r2}$  that maximizes  $\text{EU}(s)$  over the round-2 schools the student ranked, ties broken toward the school ranked higher on the student’s list. The *value of the second round* is  $v^{r2} = \text{EU}(s^{r2})$ , the quantity that enters Section 4 as the outside option  $v_{i,\text{out}}$ . A student who ranked none of the schools with vacant seats in round 2 has no second-round option: their optimal choice is undefined and their value of the second round is 0. We compute this value for all 431 survey respondents, as a counterfactual outside option for those who were in fact placed in round 1.

**Ex ante admission probabilities in the second round (Table A1).** For each school with vacant seats in the second round, we can estimate the chance that a second-round applicant is admitted. This is the second-round analogue of the ex ante admission probabilities reported for the first round in Table 2. We set each school’s round-2 capacity to the number of students actually admitted to it in the second round, and we bootstrap the round-2 applicant pool the same way as in the first round. In each of 10,000 resamples, the round-2 seats are shared equally among the resampled applicants; no second-round participant has sibling priority, so no priority step applies. Table A1 reports the resulting ex ante admission probabilities. The second round is competitive at three schools. Leidsche Rijn College attracts 37 applicants for 27 seats and admits an applicant with probability 0.74; Amadeus Lyceum has 7 applicants for 7 seats and a chance of 0.90. De Passie has 1 applicant for 1 seat but a chance of only 0.85, because a resample can draw the single applicant more than once. St. Gregorius and UniC, which carry the most residual seats, admit every applicant with probability one, and the “Outside Utrecht” fallback is unconstrained by construction. At all but these three schools, a second-round application is a sure placement.

Table A1: Ex ante (second-round) admission probabilities

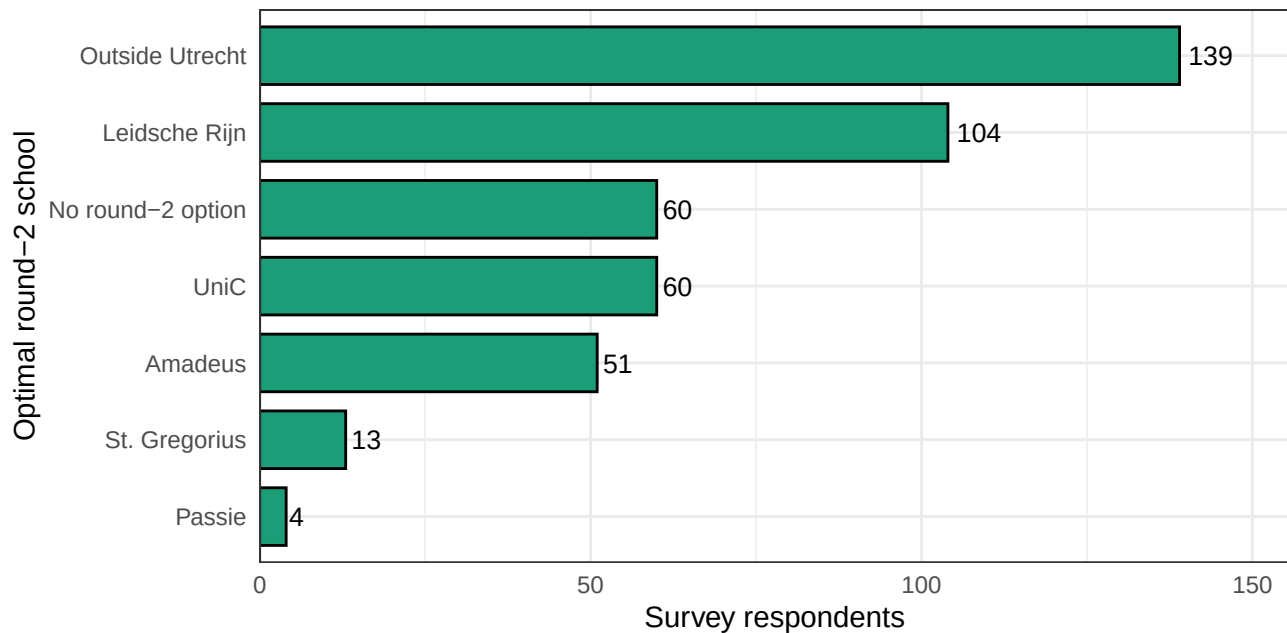
| School          | Round-2<br>capacity<br>(1) | Round-2<br>applicants<br>(2) | Admission<br>chance<br>(3) |
|-----------------|----------------------------|------------------------------|----------------------------|
| Outside Utrecht | 1,000                      | 29                           | 1.00                       |
| St. Gregorius   | 54                         | 5                            | 1.00                       |
| Leidsche Rijn   | 27                         | 37                           | 0.74                       |
| UniC            | 25                         | 14                           | 1.00                       |
| Amadeus         | 7                          | 7                            | 0.90                       |
| Passie          | 1                          | 1                            | 0.85                       |

*Notes:* The admission probability in column (3) is the mean, over the 10,000 bootstrap resamples of the round-2 applicant pool, of  $\min(1, \text{capacity}_k^2 / n_k)$ , with  $n_k$  school  $k$ ’s resampled applicant count. Round-2 applicants are the 93 students who submit a second-round list; their round-2 application is their second-round first choice. Schools with no round-2 seats are omitted.

**The optimal second-round choice (Figure A4).** Of the 431 respondents, 371 rank at least one school with vacant seats in the second round and therefore have a well-defined second-round option; the remaining 60 rank none and obtain a value of 0. Among those with an option, the optimal second-round choice is concentrated on the “Outside Utrecht” fallback (139 students) and Leidsche Rijn College (104 students), followed by UniC (60), Amadeus

Lyceum (51), St. Gregorius (13), and De Passie (4). The optimal choice trades off a school’s utility against its admission probability; Leidsche Rijn College is often optimal despite being mildly rationed (admission probability 0.74) because many students rank it highly.

Figure A4: Optimal choice in the second round

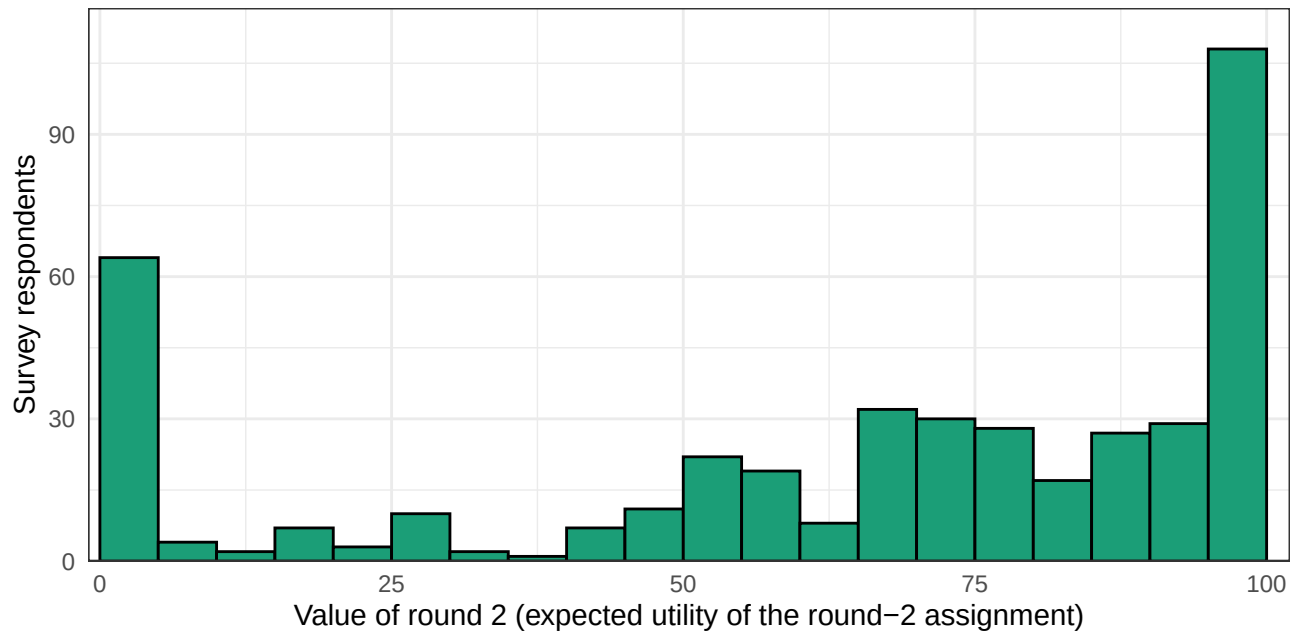


*Notes:* The figure shows, for the 431 survey respondents, the school that maximizes the expected utility of a second-round application according to equation (A1). “No round-2 option” denotes respondents who ranked none of the schools with vacant seats in the second round. Schools outside the core eleven are pooled into “Outside Utrecht”.

**The value of the second round (Figure A5).** Figure A5 plots the distribution of the value of the second round across respondents. The 60 respondents with no ranked round-2 school have a value of 0. Among the 371 respondents with a second-round option the value averages 75.7 points on the 0–100 preference-point scale, and the overall mean value of the second round is 65.1 points. The second round is thus a meaningful outside option for the large majority of respondents, which limits how costly a rejected first-round application can be. There is substantial mass at the top of the distribution: 114 respondents (26%) have a round-2 value of 95 points or more, 73 of them exactly 100. Three groups produce this mass. The first group rates an uncongested round-2 school at the top: UniC and St. Gregorius admit round-2 applicants with certainty, Amadeus Lyceum with probability 0.90. The second group rates a school outside Utrecht at or near 100 points; the outside-Utrecht alternative is unconstrained

in round 2 (60 of the 114). The third group used the option in the elicitation to award 100 points to several schools, one of which is open in round 2. For this quarter of the respondents a rejected first-round application costs almost nothing, which is the individual-level counterpart of the finding in Section 5.3 that a safe application yields little extra value. The unrated outside-Utrecht alternative is the most debatable ingredient of this construction. Appendix E therefore reruns the classification with that alternative removed from round 2 or its value cut in half. The mistake count rises to 98 and 88, and the optimization-error share stays at its baseline 62%.

Figure A5: Distribution of the value of the second round



*Notes:* The figure shows the distribution across the 431 survey respondents of the value of the second round, defined as the expected utility  $EU(s^{r2})$  of the optimal second-round application in equation (A1), measured in preference points (0–100). Respondents who ranked no school with vacant seats in the second round are assigned a value of 0.

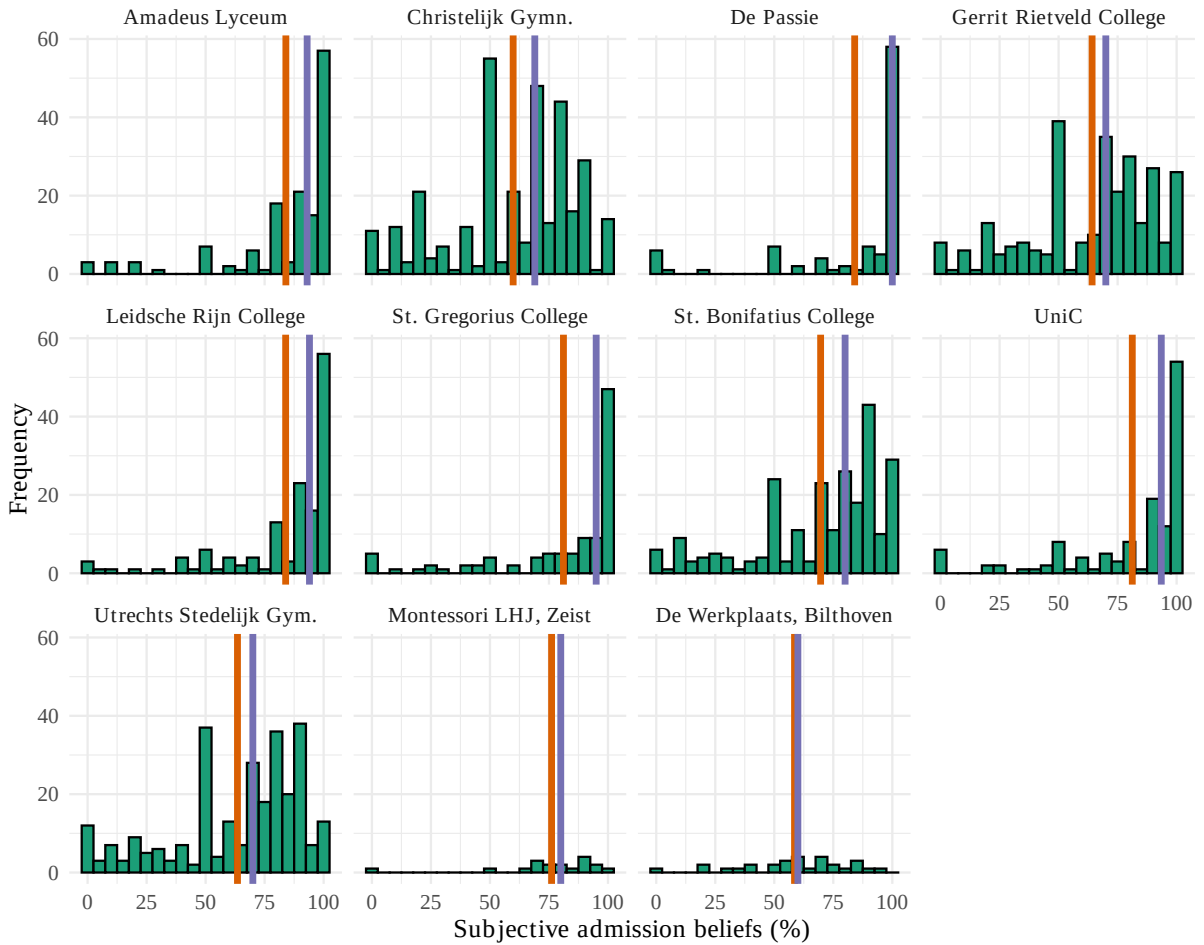
## B Additional figures and tables

Figure B1 plots for each school the distribution of students' subjective rejection probabilities.

Figures B2 and B3 in this appendix repeat Figures 1 and 2 from the main text. The figures in the main text are based on ex ante admission probabilities. The figures in this appendix are based on interim admission probabilities.

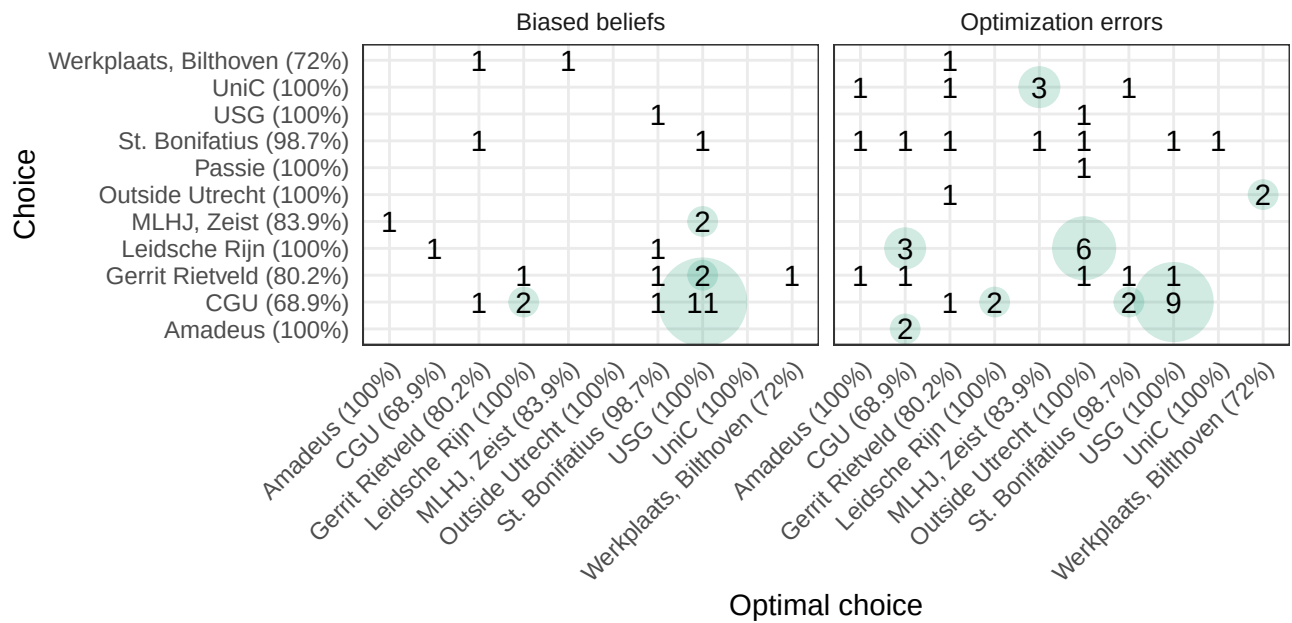
Table B1 reports for each school the number of first-round applicants, among all students and among survey respondents, and the full and survey capacities. Table B2 reports the distributional outcomes by household income discussed in Section 5.3.

Figure B1: Distribution of subjective rejection probabilities by school



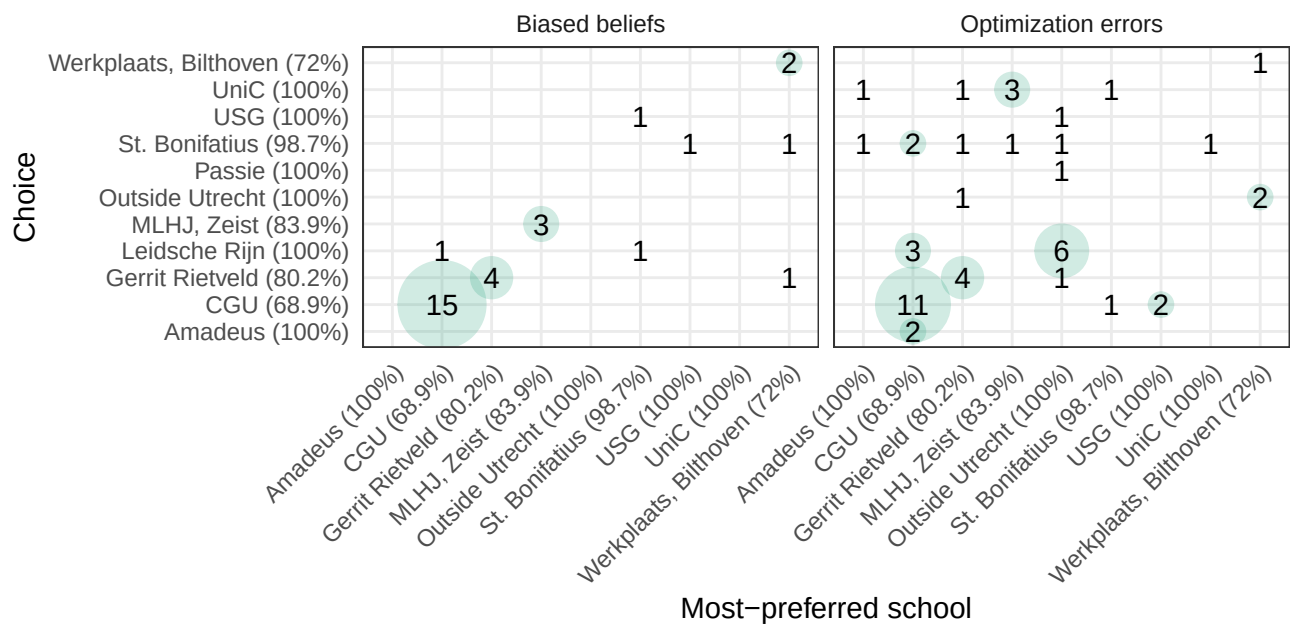
Notes: The graphs plot the distributions of subjective rejection probabilities by school. The red (blue) lines indicate the means (medians) of the distributions.

Figure B2: Strategic mistakes: choice vs. optimal choice – interim admission probabilities



*Notes:* The graphs plot all students who made a strategic mistake. On the vertical axis are the schools they actually chose and on the horizontal axes are the schools they should optimally have chosen given their preference points and the ex ante admission probabilities. Interim admission probabilities are in parentheses. The left-hand graph shows all students who made a mistake due to biased beliefs, the right-hand graph shows all students who made a mistake due to optimization errors. Numbers refer to numbers of students at specific choice/optimal choice combinations. The green circles scale with the number of students in a cell, to guide the eye toward the largest clusters.

Figure B3: Strategic mistakes: choice vs. most-preferred school – interim admission probabilities



Notes: The graphs plot all students who made a strategic mistake. On the vertical axis are the schools they actually chose and on the horizontal axes are the schools they preferred most. Interim admission probabilities are in parentheses. The left-hand graph shows all students who made a mistake due to biased beliefs, the right-hand graph shows all students who made a mistake due to optimization errors. Numbers refer to numbers of students at specific choice/most-preferred school combinations. The green circles scale with the number of students in a cell.

Table B1: Applicants and capacities by school

|                       | All<br>applicants<br>(1) | Respondents<br>applicants<br>(2) | Full<br>capacity<br>(3) | Survey<br>capacity<br>(4) |
|-----------------------|--------------------------|----------------------------------|-------------------------|---------------------------|
| Amadeus               | 69                       | 28                               | 96                      | 39                        |
| CGU                   | 190                      | 95                               | 170                     | 71                        |
| Passie                | 10                       | 3                                | 25                      | 8                         |
| Gerrit Rietveld       | 110                      | 63                               | 93                      | 53                        |
| Leidsche Rijn         | 108                      | 42                               | 138                     | 54                        |
| St. Gregorius         | 6                        | 2                                | 60                      | 20                        |
| St. Bonifatius        | 155                      | 63                               | 153                     | 62                        |
| UniC                  | 42                       | 21                               | 72                      | 36                        |
| USG                   | 77                       | 38                               | 166                     | 38                        |
| MLHJ, Zeist           | 56                       | 19                               | 47                      | 16                        |
| Werkplaats, Bilthoven | 25                       | 13                               | 18                      | 9                         |
| Outside Utrecht       | 102                      | 44                               | 1,000                   | 1,000                     |
| Total                 | 950                      | 431                              | 2,038                   | 1,406                     |

Notes: The survey capacity in column (4) is  $\text{capacity}_k^{\text{survey}} = \text{round}\left(\text{capacity}_k \times n_k^{\text{survey}} / (n_k^{\text{all}} + n_k^{\text{ext}})\right)$ , where  $n_k^{\text{ext}}$  is the number of out-of-district applicants who compete for seats at school  $k$  but are not in our register data (39 at CGU, 89 at USG, zero elsewhere). The “Outside Utrecht” row collects students whose first-round choice is a school outside the Utrecht choice set; it carries a nominal capacity of 1,000 in both samples.

Table B2: Distributional outcomes by household income, by allocation

|                                 | Dissimilarity<br>index<br>(1) | Gymnasium share (%) |               |
|---------------------------------|-------------------------------|---------------------|---------------|
|                                 |                               | Low-income<br>(2)   | Other<br>(3)  |
| Actual Boston                   | 0.169<br>(0.055)              | 25.6<br>(6.1)       | 25.3<br>(2.3) |
| Corrected beliefs (equilibrium) | 0.167<br>(0.056)              | 26.6<br>(6.3)       | 25.1<br>(2.3) |
| Optimal choices (equilibrium)   | 0.149<br>(0.053)              | 24.5<br>(6.1)       | 25.4<br>(2.3) |
| DA-STB                          | 0.132<br>(0.050)              | 25.3<br>(6.1)       | 25.3<br>(2.3) |

Notes: Low-income denotes the 50 students from below-median-income households; other, the remaining 381 respondents. The dissimilarity index is  $\frac{1}{2} \sum_k |\ell_k / L - o_k / O|$ , where  $\ell_k$  and  $o_k$  are the low-income and other students placed at school  $k$ ; it is 0 when every school has the sample low-income share and 1 under complete separation, and counts placement outside the Utrecht choice set as a category (restricting to the Utrecht schools leaves the pattern unchanged). The gymnasium shares are shares placed at Christelijk Gymnasium Utrecht or Utrechts Stedelijk Gymnasium. All entries are means over 1,000 lottery draws, with common random numbers across allocations. Standard errors, in parentheses, are from a student bootstrap (2,000 resamples) and capture sampling uncertainty in the student pool, holding the market fixed.

## C The strategic-mistake classification in a graph

This appendix gives a graphical representation of the strategic-mistake classification of Section 4 and of its decomposition into biased-belief mistakes and optimization errors, summarized in Section 5.1. The figures use the primitives of the main text: the chosen school  $s_i^*$ , the preference points  $v_{ik}$ , the outside option  $v_{i,\text{out}} = v_i^2$ , the admission probabilities  $\alpha_{ik}$  (Section 4), and the subjective admission beliefs (Section 3). The only new object is the binding alternative  $q_i^*$ , the school against which the chosen school is compared. For mistake-makers,  $q_i^*$  is the school that maximizes the expected utility of Section 4. For the other students, it is the expected-utility maximizer among the schools they did not apply to, the closest call they rejected. Ties are resolved by preferring the school ranked higher in the student’s rank-ordered list.

The strategic-mistake inequality of Section 4 can then be rewritten as a pairwise comparison between the chosen school and the binding alternative. Each student is a single point on the plane defined by

$$x_i = \frac{v_{is_i^*} - v_{i,\text{out}}}{v_{iq_i^*} - v_{i,\text{out}}}, \quad y_i = \frac{\alpha_{is_i^*}}{\alpha_{iq_i^*}}.$$

Both ratios are normalized by the outside option:  $x_i$  is the *relative utility gain* of the chosen school over the binding alternative, both measured net of the safe fallback;  $y_i$  is the *relative admission probability* of the chosen school over the binding alternative. In this parameterization, the no-mistake condition becomes  $x_i \cdot y_i \geq 1$ : points on or above the rectangular hyperbola  $y = 1/x$  correspond to students who did not make a strategic mistake, points strictly below correspond to mistakes.<sup>19</sup>

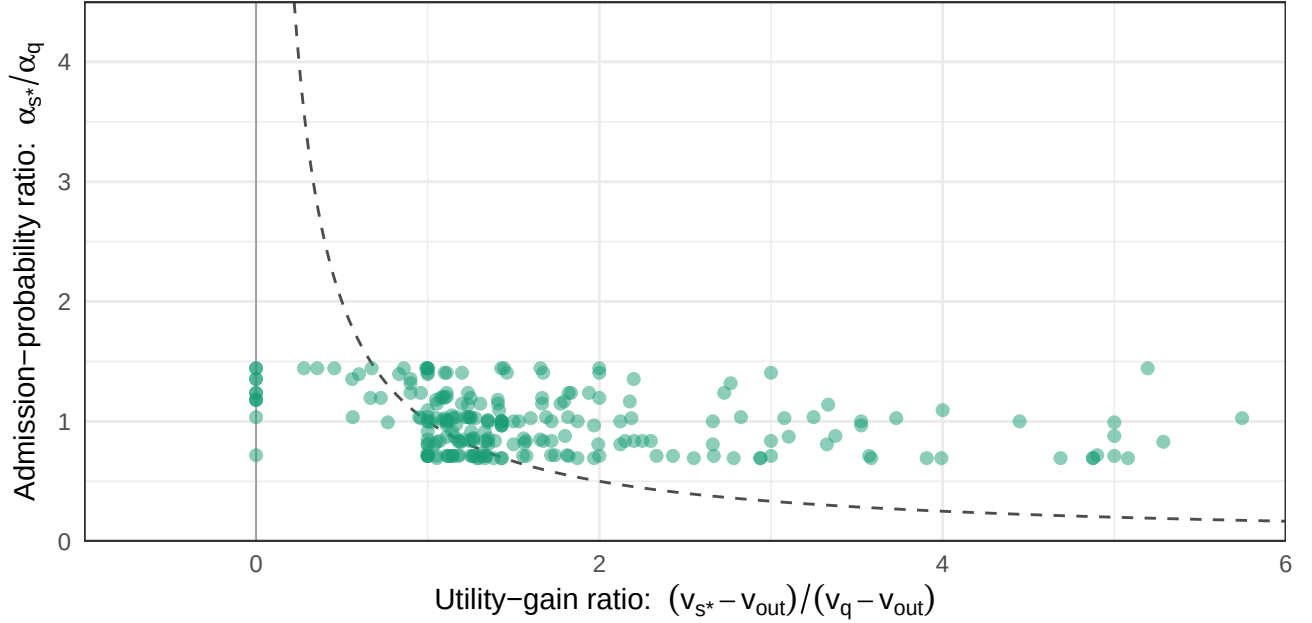
A few students sit at boundaries of the plane. Students whose binding alternative is weakly dominated by the outside option ( $v_{iq_i^*} \leq v_{i,\text{out}}$ ) have no well-defined  $x_i$  and are excluded from the plots; because the round-2 value exceeds the binding alternative for many students, these exclusions are the most common case, and the figure notes report the counts. Students who chose a school worse than their own outside option ( $v_{is_i^*} < v_{i,\text{out}}$ ) have  $x_i < 0$  and are mistake-makers by definition; they appear to the left of the vertical line at  $x = 0$ . One student is convinced she has no chance at the binding alternative, so  $y_i$  is undefined under subjective beliefs and she is excluded from Figure C2. Students exactly on the curve are classified by the tie-breaking rule: ties go to the school ranked higher on the student’s list.

<sup>19</sup>To keep the ratios well defined under subjective beliefs we assign an admission probability of 1% to students who report a 0% admission probability at the school they actually applied to. When a student’s highest reported admission probability across the in-Utrecht schools is below one, the “Outside Utrecht” option is reassigned a subjective probability of one.

The plots clip to  $x \in [-1, 6]$  and  $y \in [0, 4.5]$ ; the notes report the points outside the window.

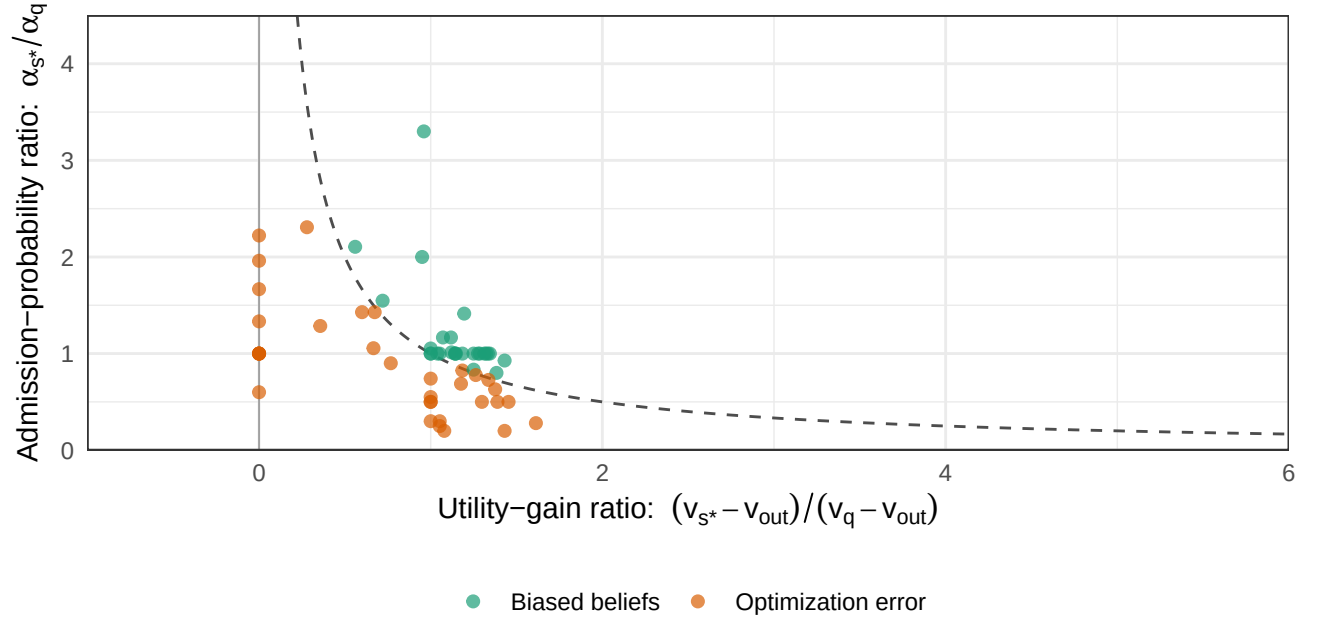
Figure C1 plots all students for whom the binding alternative is not weakly dominated by the outside option, using ex ante admission probabilities. Figure C2 replots the mistake-makers using their *subjective* admission beliefs in place of the rational ones. The position of each mistake-maker in Figure C2 then has an interpretation: mistake-makers who lie *above* the curve under subjective beliefs are those whose mistake is classified as biased beliefs (the inequality would hold if their beliefs were correct), while mistake-makers who remain *below* the curve are those whose mistake persists even given their own beliefs, and so is classified as an optimization error.

Figure C1: Strategic-mistake inequality – ex ante admission probabilities, all students



Notes: Each point is one student. Horizontal axis: utility-gain ratio  $(v_{s^*} - v_{out}) / (v_q - v_{out})$  for the binding alternative  $q$ . Vertical axis: admission-probability ratio  $\alpha_{s^*} / \alpha_q$ . The dashed curve is  $y = 1/x$ ; the vertical gray line at  $x = 0$  marks the locus where the chosen school is no better than the outside option. Points on or above the curve correspond to students whose application choice satisfies the strategic-mistake inequality given ex ante admission probabilities; points below the curve correspond to mistake-makers; points to the left of  $x = 0$  chose a school worse than their outside option and are mistake-makers by definition. Of the 78 mistake-makers reported in Section 5.1, 65 lie strictly below the curve and 13 are excluded because the binding alternative is weakly dominated by the outside option, so the hyperbola representation is undefined for them (their classification follows from the direct expected-utility comparison). Of the 353 non-mistake-makers, 165 are excluded for the same reason and 4 have no binding alternative, leaving 184 plotted; 249 students are plotted in total. The plot window is clipped to  $x \in [-1, 6]$  and  $y \in [0, 4.5]$ ; 18 of 249 plotted students (7.2%) lie outside this window and are not shown.

Figure C2: Strategic-mistake inequality – subjective admission beliefs, mistake-makers only



*Notes:* Each point is one of the 78 mistake-makers of Section 5.1, replotted using subjective admission beliefs in place of ex ante probabilities; 64 have well-defined coordinates and are shown (the 14 excluded students, all optimization errors, have a binding alternative weakly dominated by the outside option, and their classification follows from the direct expected-utility comparison). The dashed curve is  $y = 1/x$ ; the vertical gray line at  $x = 0$  marks the locus where the chosen school is no better than the outside option. Points above the curve are mistakes classified as biased beliefs (the inequality holds under the student's own beliefs); points below or to the left of  $x = 0$  are mistakes classified as optimization errors (the inequality fails even under the student's own beliefs). The students lying exactly on the curve are resolved by the tie-breaking rule based on the rank-ordered list. The plot window is clipped to  $x \in [-1, 6]$  and  $y \in [0, 4.5]$ ; 3 of 64 plotted mistake-makers (4.7%) lie outside this window and are not shown.

## D Results based on estimated instead of elicited cardinal preferences

The results in the main text are based on cardinal preferences elicited through the survey question that asks students to award preference points to schools relative to the first-ranked school, which receives 100 points. This appendix derives the main results using cardinal utilities estimated from a rank-ordered logit model based on students’ submitted rank-ordered lists.

**Estimation.** We estimate a rank-ordered (exploded) logit (Beggs et al., 1981) on the 431 respondents’ rankings of the twelve analysis schools (the eleven Utrecht-area schools and the composite “outside Utrecht” alternative). The deterministic part of the utility of student  $i$  for school  $s$  is

$$V_{is} = \gamma \text{distance}_{is} + \beta (\text{test score}_i \times \text{gymnasium}_s) + \eta_s,$$

where  $\eta_s$  are school fixed effects,  $\text{distance}_{is}$  is the home–school distance, and  $\text{gymnasium}_s$  indicates the two single-track gymnasia (CGU and USG). Schools the student did not rank are treated as unacceptable, i.e. ranked below every ranked school. The estimates are intuitive: students dislike distance ( $\hat{\gamma} = -0.31$ ) and higher-ability students are more likely to prefer the gymnasia ( $\hat{\beta} = 0.68$ ), both highly significant.

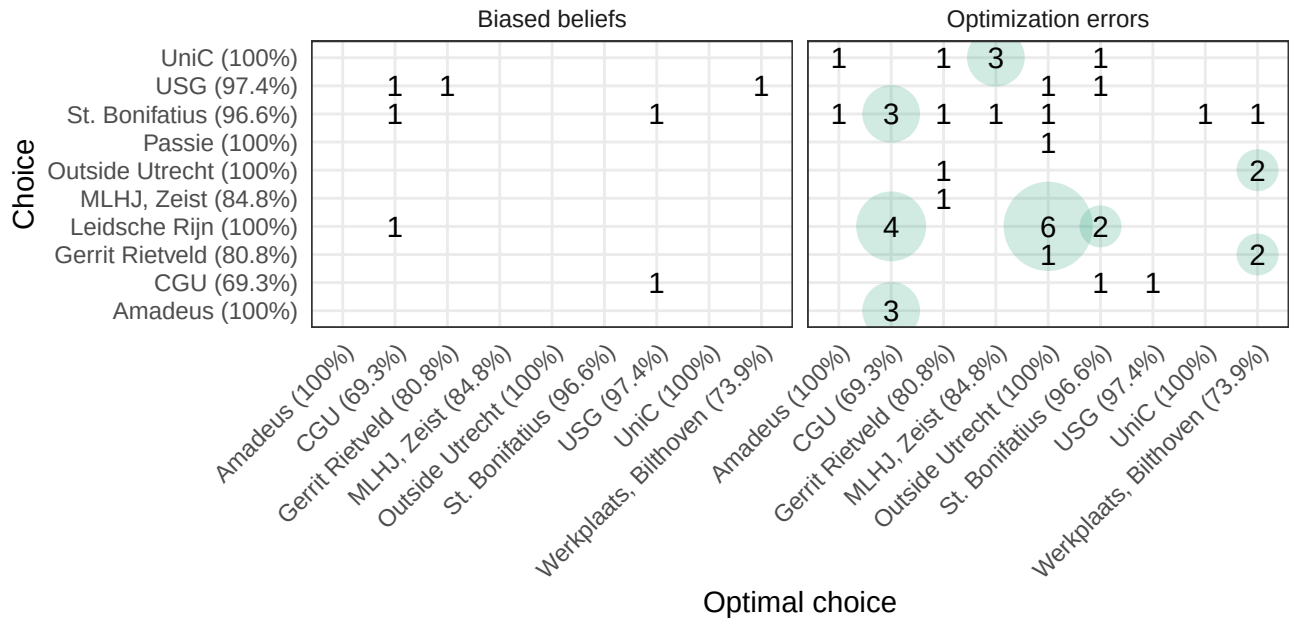
Given the estimated  $V_{is}$ , we recover total utilities  $U_{is} = V_{is} + \varepsilon_{is}$ , with  $\varepsilon_{is}$  i.i.d. Type-I extreme value, by sampling utilities consistent with each student’s observed ranking. We use the classical equivalence between the rank-ordered logit and a race of exponential arrival times: because  $\exp(-U_{is})$  is exponentially distributed with rate  $\exp(V_{is})$ , ranking schools by utility is equivalent to ordering these arrival times (Yellott, 1977). We draw arrival times consistent with each student’s ranking, using the representation of exponential order statistics as a sequence of independent exponential increments (Rényi, 1953) (ranked schools in the submitted order, every unranked school strictly later than the lowest-ranked school), and set  $U_{is} = -\log(\text{arrival time})$ . Averaging over 5,000 draws gives each student’s estimated cardinal utility for each school. By construction every student’s most-preferred school has the highest estimated utility and every unranked school lies below the lowest-ranked one. The estimated utilities are mapped to the points scale of the main text by a single global linear transform (preserving all cross-student cardinal differences); the strategic-mistake classification is invariant to this transform. The estimated utilities correlate 0.66 with the elicited

preference points over ranked schools.

The outside option is the value of the second round of Section 4, recomputed with the estimated utilities: the same round-2 admission probabilities, the same ranked round-2 schools, and the same safe fallback, so that this appendix varies only the utility measure.

**Strategic mistakes.** Figures D1 and D2 repeat the main-text Figures 1 and 2 using the estimated utilities. Under the estimated measure 49 of 431 students (11.4%) make a strategic mistake, of which 7 are due to biased beliefs and 42 to optimization errors. This is fewer than the 78 (18.1%; 29 belief, 49 optimization) found with the elicited preference points, but the pattern is the same. Optimization errors dominate even more strongly, at 86% of the mistakes.

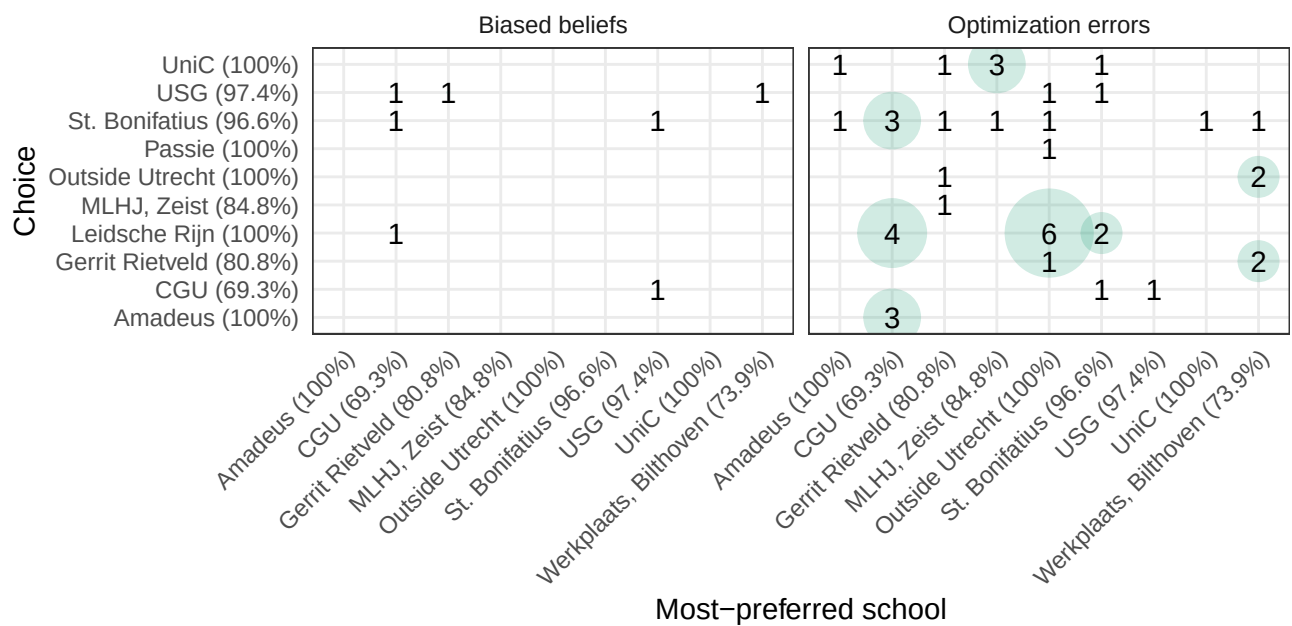
Figure D1: Strategic mistakes: choice vs. optimal choice – estimated cardinal preferences



Notes: As Figure 1, but cardinal utilities are estimated from a rank-ordered logit on students' rank-ordered lists rather than elicited preference points. Ex ante admission probabilities are in parentheses.

**Why fewer mistakes?** The drop from 78 to 49 mistakes has a single dominant source. A choice is a mistake when, relative to the binding alternative  $q$ , the point  $(x, y)$  with  $x = (v_{s^*} - v_{\text{out}})/(v_q - v_{\text{out}})$  and  $y = \alpha_{s^*}/\alpha_q$  lies below the curve  $y = 1/x$  (Section 4 and Appendix C). One observation identifies the mechanism. The ratio  $y$  does not depend on the

Figure D2: Strategic mistakes: choice vs. most-preferred school – estimated cardinal preferences



Notes: As Figure 2, but using estimated rather than elicited cardinal preferences.

utility measure: admission probabilities are a function of schools and sibling priority, not of preferences, so  $y$  is identical under the elicited and estimated measures. The difference thus operates through  $x$ , the intensity with which a student prefers their chosen school over the binding alternative. Holding the binding alternative fixed, all 39 students who made a mistake under the elicited measure but not under the estimated one move from below to above the curve because  $x$  rises. Their median  $x$  increases from 1.15 to 1.90, while  $y$  stays at 0.71 (Figure D3). The rank-ordered-logit utilities thus imply a larger preference premium for the chosen school over the next-best alternative than the elicited points do, so the same sacrifice in admission probability is more often worth it. A smaller group of 10 students becomes a mistake only under the estimated measure, which leaves the net count at 49.

This larger premium is visible directly in the cardinal utilities by rank. Table D1 reports, on a scale that sets each student’s most-preferred school to 100 and their unranked schools to 0, the average utility of the school at each position of the preference ordering. The drop from the first to the second choice is much steeper under the estimated measure ( $100 \rightarrow 63$ ) than under the elicited points ( $100 \rightarrow 83$ ). On the elicited measure the second-favorite school is worth more than four fifths of the favorite, while the rank-ordered logit infers a sharper fall-off at the top. Because the binding alternative for most of these students is a higher-ranked but safer school, this steeper top makes the preference-intensity ratio  $x$  larger and pushes them above the curve.

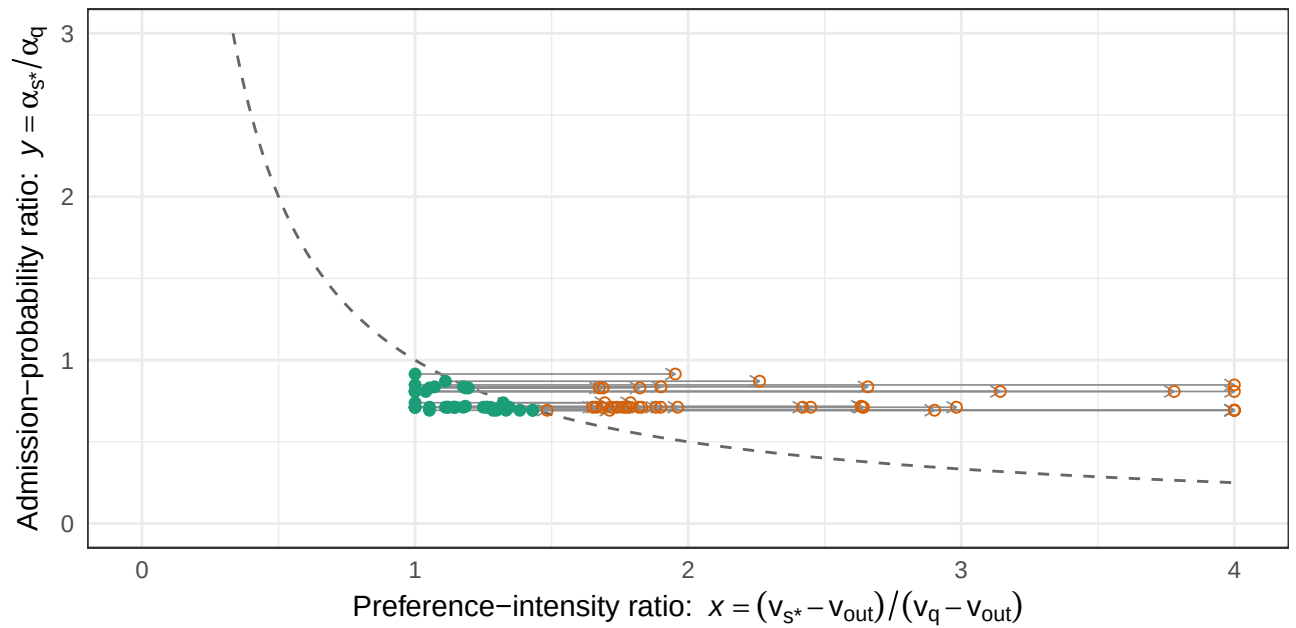
Table D1: Mean cardinal utility by rank: elicited versus estimated

|        | Elicited | Estimated |
|--------|----------|-----------|
| Rank 1 | 100      | 100       |
| Rank 2 | 83       | 63        |
| Rank 3 | 66       | 45        |
| Rank 4 | 53       | 31        |
| Rank 5 | 44       | 20        |

*Notes:* Utilities are normalized per student so that the most-preferred school equals 100 and the unranked schools equal 0. For the elicited measure this is the raw preference-points scale; for the estimated measure the zero anchor is the mean estimated utility of the student’s unranked schools.

**Welfare losses and counterfactuals.** Tables D2 and D3 repeat the welfare-loss and mechanism-comparison analyses with the estimated utilities. The conclusions are robust. Optimization errors carry larger welfare losses than biased-belief mistakes. Correcting beliefs alone has a negligible aggregate effect. As in the main text, removing all strategic mistakes does not give Boston a significant advantage over DA-STB. Under the estimated measure the optimal-

Figure D3: Why the estimated measure finds fewer mistakes: the 39 students whose classification changes



Notes: Each arrow is one of the 39 students classified as a strategic mistake under the elicited preference points but not under the estimated rank-ordered-logit utilities, with the binding alternative held fixed at the elicited optimal school. Filled (green) points use the elicited measure; open (orange) points use the estimated measure. The dashed curve is  $y = 1/x$ ; points below it are mistakes. Because the admission-probability ratio  $y$  does not depend on the utility measure, every arrow is horizontal: the students cross the curve solely because the preference-intensity ratio  $x$  increases under the estimated utilities.

choices contrast is +0.74 points (s.e. 0.47,  $p = 0.12$ ) in equilibrium, against  $-0.21$  ( $p = 0.47$ ) under the elicited points. The point estimates differ in sign but agree in substance. Under optimal play the two mechanisms are statistically indistinguishable under either preference measure. Welfare here is expressed in the estimated-utility units, so the levels are not directly comparable to the elicited tables.

Table D2: Welfare loss by mistake type – estimated cardinal preferences

|                     | Biased beliefs<br>(1) | Opt. error<br>(2) | All mistakes<br>(3) |
|---------------------|-----------------------|-------------------|---------------------|
| Mean (pref. points) | 31.65                 | 50.03             | 47.40               |
| SD                  | (12.83)               | (23.47)           | (23.09)             |
| p10                 | 19.45                 | 30.72             | 23.49               |
| Median              | 28.15                 | 46.04             | 45.70               |
| p90                 | 47.85                 | 91.74             | 78.17               |
| # Students          | 7                     | 42                | 49                  |

*Notes:* p10 and p90 denote the 10th and 90th percentiles of the loss distribution; standard deviations in parentheses.

Table D3: Mean welfare by mechanism and mistake type – estimated cardinal preferences

|                                                                  | All<br>(1)         | No mistake<br>(2)  | Biased beliefs<br>(3) | Opt. error<br>(4)   |
|------------------------------------------------------------------|--------------------|--------------------|-----------------------|---------------------|
| <i>Panel A: Mean welfare (estimated utilities, points scale)</i> |                    |                    |                       |                     |
| Actual Boston                                                    | 84.83<br>(1.55)    | 90.89<br>(1.42)    | 41.69<br>(9.53)       | 36.86<br>(3.93)     |
| Corrected beliefs (equilibrium)                                  | 85.20<br>(1.54)    | 90.78<br>(1.44)    | 69.75<br>(9.35)       | 37.08<br>(3.93)     |
| Optimal choices (equilibrium)                                    | 88.27<br>(1.37)    | 89.52<br>(1.47)    | 67.63<br>(10.14)      | 80.27<br>(3.25)     |
| DA-STB                                                           | 87.53<br>(1.25)    | 88.34<br>(1.37)    | 68.52<br>(8.37)       | 83.35<br>(2.39)     |
| Constrained-efficient (ceiling)                                  | 94.26<br>(1.30)    | 95.57<br>(1.37)    | 77.42<br>(11.68)      | 85.15<br>(4.13)     |
| <i>Panel B: Change from actual Boston</i>                        |                    |                    |                       |                     |
| Corrected beliefs (equilibrium)                                  | +0.37<br>(0.24)    | -0.12<br>(0.11)    | +28.05**<br>(9.13)    | +0.22<br>(0.14)     |
| Optimal choices (equilibrium)                                    | +3.44***<br>(0.82) | -1.37***<br>(0.22) | +25.94**<br>(10.37)   | +43.41***<br>(4.39) |
| DA-STB                                                           | +2.70***<br>(0.86) | -2.56***<br>(0.39) | +26.83***<br>(5.42)   | +46.49***<br>(3.23) |
| <i>Panel C: Difference from DA-STB</i>                           |                    |                    |                       |                     |
| Optimal choices (equilibrium) – DA-STB                           | +0.74<br>(0.47)    | +1.18**<br>(0.47)  | -0.89<br>(5.14)       | -3.08<br>(2.14)     |
| # Students                                                       | 431                | 382                | 7                     | 42                  |

*Notes:* The table compares mean student welfare across four scenarios, simulated for the 431 survey respondents with school capacities scaled to the survey sample (Table B1), sibling priority, and a single lottery number per student per draw (1,000 draws). Scenarios as defined in the text: in the correction scenarios the 7 biased-belief mistake-makers (respectively all 49 mistake-makers) switch to their optimal first choice in the Nash equilibrium of the application game; under DA-STB each student submits their full stated ranking. Welfare is the mean estimated utility of the assigned school across draws, mapped to the points scale by a single global linear transform. The constrained-efficient ceiling maximizes total preference points subject to capacities (full cardinal information, no priorities or lotteries). Panel C reports the within-student difference between the optimal-choices Boston counterfactuals and DA-STB. Standard errors, in parentheses, are cross-student sampling errors, not simulation errors: in Panel A they are computed across students within each column, and in Panels B and C from the within-student (paired) differences. Stars denote significance from a within-student  $t$ -test: \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## E Sensitivity of the classification

This appendix probes the strategic-mistake classification along three dimensions: measurement noise in the elicited preference points, the specification of the round-2 outside option, and the distance of individual students to the classification boundary.

**Noise in the elicited preference points.** A mistake is classified as an optimization error whenever the stated beliefs cannot rationalize it, so “optimization error” is a residual category. Noise in the elicited points that makes a correct choice look suboptimal creates a spurious mistake. Because noisy points are unlikely to be consistent with the separately stated beliefs, the spurious mistake also tends to land in the optimization-error category. Table E1 quantifies this channel. In each of 200 simulation draws per noise level, we add mean-zero normal noise to the elicited points of every ranked school below the top rank. The top school stays at 100 by normalization. We rebuild the round-2 outside option from the noisy points and rerun the full classification. Noise indeed inflates the mistake count and loads the additions onto the optimization-error side. But the effect is modest. At a noise standard deviation of 5 points, roughly a quarter of the mean gap between first- and second-ranked schools, the mean count is 78.7 against 78 at baseline. The optimization-error share of mistakes and of welfare loss stays roughly constant.

**The round-2 outside option.** The outside option  $v_{i,\text{out}}$  enters every classification inequality. Section 4 reconstructs it from a second round whose actual assignment procedure is not documented. Table E2 reruns the classification under eleven perturbations of this reconstruction: direct shifts of the round-2 value ( $\times 0.75$ ,  $\times 1.25$ ,  $\pm 10$  points), shifted round-2 admission probabilities ( $\pm 0.10$ , and all round-2 schools safe), alternative fallback rules after a round-2 rejection (no fallback, and the best ranked school regardless of congestion), and two variants that target the composite outside-Utrecht alternative. The baseline offers this alternative in round 2 without rationing at the student’s stated value. The two variants remove it from the round-2 option set entirely or cut its stated value in half. The mistake count moves between 65 and 98 across these variants, against 78 at baseline. Optimization errors remain the majority of mistakes (58–74%) in every one. In the two outside-Utrecht variants their share is 62%, exactly the baseline value. The welfare comparison of Section 5.3 is affected only through which choices are deemed optimal, since the simulations re-run the second round in every draw. Re-running the optimal-choices counterfactual for one variant from each perturbation family gives a Boston-minus-DA-STB contrast of  $+0.68$  (s.e. 0.40) at  $v^{\text{out}} \times 0.75$ ,  $-1.60$  (s.e.

Table E1: Sensitivity of the classification to noise in the elicited points

|                           | Mistake counts   |                |                  | Opt. error share (%) |                   | Opt. majority (% draws) |                   |
|---------------------------|------------------|----------------|------------------|----------------------|-------------------|-------------------------|-------------------|
|                           | All<br>(1)       | Beliefs<br>(2) | Opt. err.<br>(3) | Mistakes<br>(4)      | Welf. loss<br>(5) | Mistakes<br>(6)         | Welf. loss<br>(7) |
| Baseline ( $\sigma = 0$ ) | 78               | 29             | 49               | 62.8                 | 84.9              | –                       | –                 |
| $\sigma = 2.5$            | 76.0<br>[73, 80] | 28.0           | 48.1             | 63.3                 | 84.5              | 100.0                   | 100.0             |
| $\sigma = 5$              | 78.7<br>[74, 84] | 27.5           | 51.2             | 65.1                 | 84.9              | 100.0                   | 100.0             |
| $\sigma = 10$             | 85.8<br>[79, 92] | 27.5           | 58.3             | 68.1                 | 85.3              | 100.0                   | 100.0             |

*Notes:* In each draw the noisy utility of a ranked school below the first ( $v = 100$  – preference points) is clipped to  $[0, 100]$ ; the first-ranked school is the scale anchor ( $v = 100$  by normalization) and is held fixed, as are unranked schools ( $v = 0$ ). The noise is propagated into the round-2 outside option, which is recomputed per draw from the noisy points. Columns (1)–(3) report the mean number of strategic mistakes and their split into biased beliefs and optimization errors across 200 draws per  $\sigma$ ; the bracketed interval under column (1) is the  $[p10, p90]$  range of the mistake count. Columns (4)–(5) give the mean share of mistakes and of the total welfare loss attributed to optimization errors; columns (6)–(7) the share of draws in which optimization errors account for more than half of the mistakes and of the welfare loss. Welfare loss is the expected-utility difference between the optimal and the chosen school under the actual admission probabilities, evaluated at the draw’s noisy points.

0.35) at  $v^{\text{out}} \times 1.25$ ,  $-0.44$  (s.e. 0.48) with no round-2 fallback, and  $+0.69$  (s.e. 0.43) with the outside-Utrecht option excluded. The contrast thus lies between  $-1.60$  and  $+0.69$  across the perturbation families, and no variant delivers a significant Boston advantage at the 5% level. The sign of the small contrast is sensitive to the outside-option scale. This is why the main text concludes that Boston has no robust advantage under optimal play, rather than leaning on the point estimate. DA-STB’s advantage over actual Boston does not depend on how the outside option is constructed.

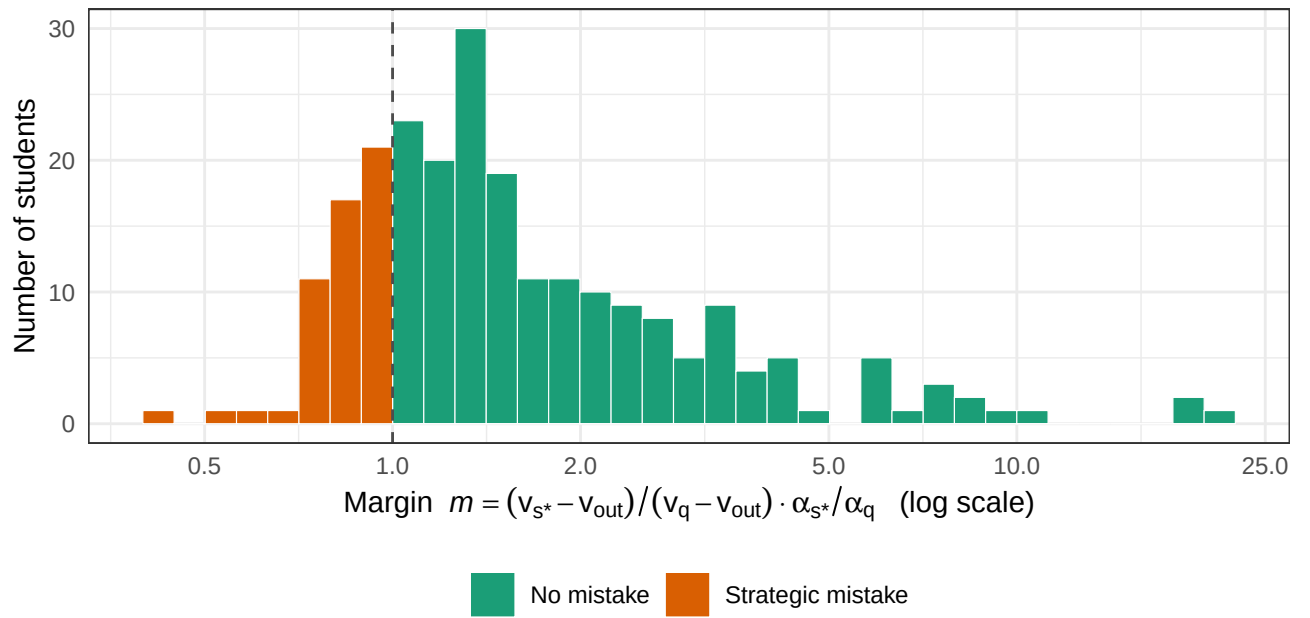
**Distance to the classification boundary.** Figure E1 shows how far students sit from the classification boundary. It uses the margin  $m = x \cdot y$  of Appendix C; a choice is optimal iff  $m \geq 1$ . Of the 65 mistake-makers with a well-defined margin, 17 (26%) sit within 10% of the boundary. They would flip to no-mistake under a 10% increase in  $m$ . Conversely, 16 of the 184 plotted no-mistake students (9%) sit within 10% on the other side. The margins line up with the welfare losses of Section 5.2: the marginal mistakes are cheap, and the costly tail sits far from the boundary. The belief/optimization classification is less fragile than the mistake margin itself. Under subjective beliefs, 6 of the 29 biased-belief labels (21%) and 4 of the 35 optimization-error labels with a defined margin (11%) sit within 10% of their boundary.

Table E2: Sensitivity of the classification to the round-2 outside option

|                                                       | Mistakes<br>(N)<br>(1) | Biased<br>beliefs (N)<br>(2) | Opt. errors<br>(N)<br>(3) | Opt. errors<br>(%)<br>(4) | Welfare<br>loss<br>(5) |
|-------------------------------------------------------|------------------------|------------------------------|---------------------------|---------------------------|------------------------|
| Baseline                                              | 78                     | 29                           | 49                        | 63                        | 13.66                  |
| <i>Panel A: Direct shifts of the round-2 value</i>    |                        |                              |                           |                           |                        |
| $v^{\text{out}} \times 0.75$                          | 93                     | 39                           | 54                        | 58                        | 12.83                  |
| $v^{\text{out}} \times 1.25$                          | 77                     | 22                           | 55                        | 71                        | 14.40                  |
| $v^{\text{out}} - 10$                                 | 83                     | 35                           | 48                        | 58                        | 13.63                  |
| $v^{\text{out}} + 10$                                 | 65                     | 17                           | 48                        | 74                        | 15.65                  |
| <i>Panel B: Round-2 admission probabilities</i>       |                        |                              |                           |                           |                        |
| $\alpha - 0.10$                                       | 86                     | 36                           | 50                        | 58                        | 13.17                  |
| $\alpha + 0.10$                                       | 71                     | 24                           | 47                        | 66                        | 14.76                  |
| All schools safe ( $\alpha = 1$ )                     | 71                     | 23                           | 48                        | 68                        | 14.60                  |
| <i>Panel C: Fallback after a round-2 rejection</i>    |                        |                              |                           |                           |                        |
| Fallback = 0                                          | 82                     | 32                           | 50                        | 61                        | 13.13                  |
| Fallback = best ranked school                         | 71                     | 23                           | 48                        | 68                        | 14.60                  |
| <i>Panel D: The outside-Utrecht option in round 2</i> |                        |                              |                           |                           |                        |
| Outside Utrecht excluded                              | 98                     | 37                           | 61                        | 62                        | 13.37                  |
| Outside-Utrecht value halved                          | 88                     | 33                           | 55                        | 62                        | 12.94                  |

Notes: Each row reclassifies strategic mistakes for the 431 survey respondents after perturbing the modeled value of the second round ( $v^{\text{out}}$ ), the outside option of the classification. Baseline:  $v^{\text{out}}$  as constructed in Section 4. Panel A shifts the baseline value directly (clipped to the 0–100 scale). Panel B shifts every round-2 admission probability (clipped to  $[0, 1]$ , congestion flags re-derived) or makes all round-2 schools safe. Panel C replaces the fallback after a round-2 rejection by 0 or by the best ranked round-2 school regardless of congestion. Panel D removes the unrationed outside-Utrecht alternative from round 2 or halves its stated value. Columns (1)–(4): strategic mistakes, their split, and the optimization-error share; column (5): mean expected-utility loss of mistake-makers under the actual admission probabilities, evaluated with the variant’s outside option. The welfare contrasts for four variants are reported in the text:  $+0.68$  (se 0.40) at  $v^{\text{out}} \times 0.75$ ,  $-1.60$  (se 0.35) at  $v^{\text{out}} \times 1.25$ ,  $-0.44$  (se 0.48) with fallback 0, and  $+0.69$  (se 0.43) with the outside-Utrecht option excluded.

Figure E1: Distance to the classification boundary



Notes: Histogram of the classification margin  $m = x \cdot y$  (Appendix C) under ex ante admission probabilities, for the 249 students with a well-defined margin; the dashed line marks the boundary  $m = 1$ , below which a choice is a strategic mistake. The horizontal axis is log-scaled and the display is clipped to  $m \in [0.2, 25]$ ; 12 students below the window (all with  $m \leq 0$ : their chosen school is worse than the outside option) and 3 above it are included in all statistics but not shown.